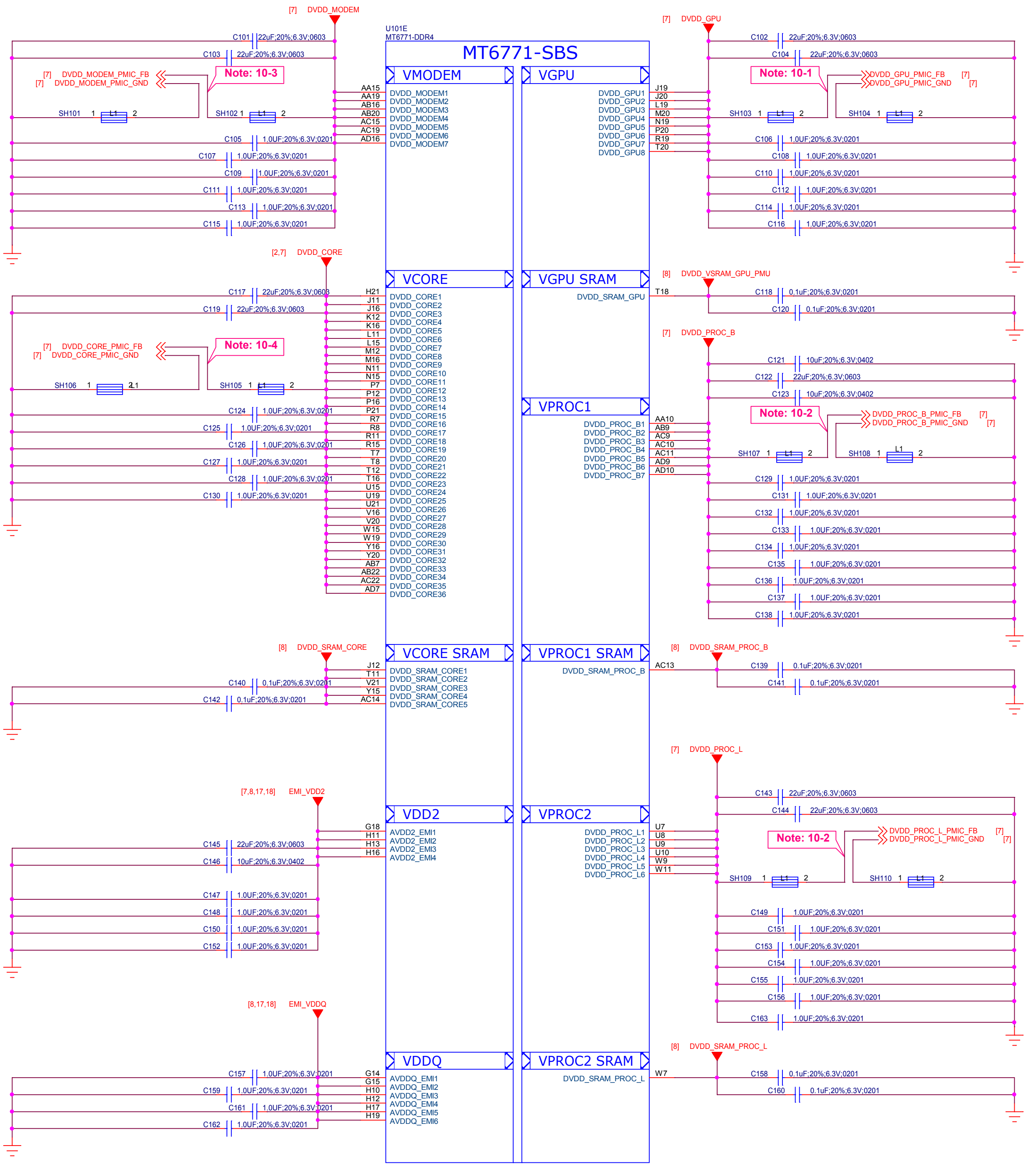


Schematic design notice of "10_BB_POWER_PDN" page.

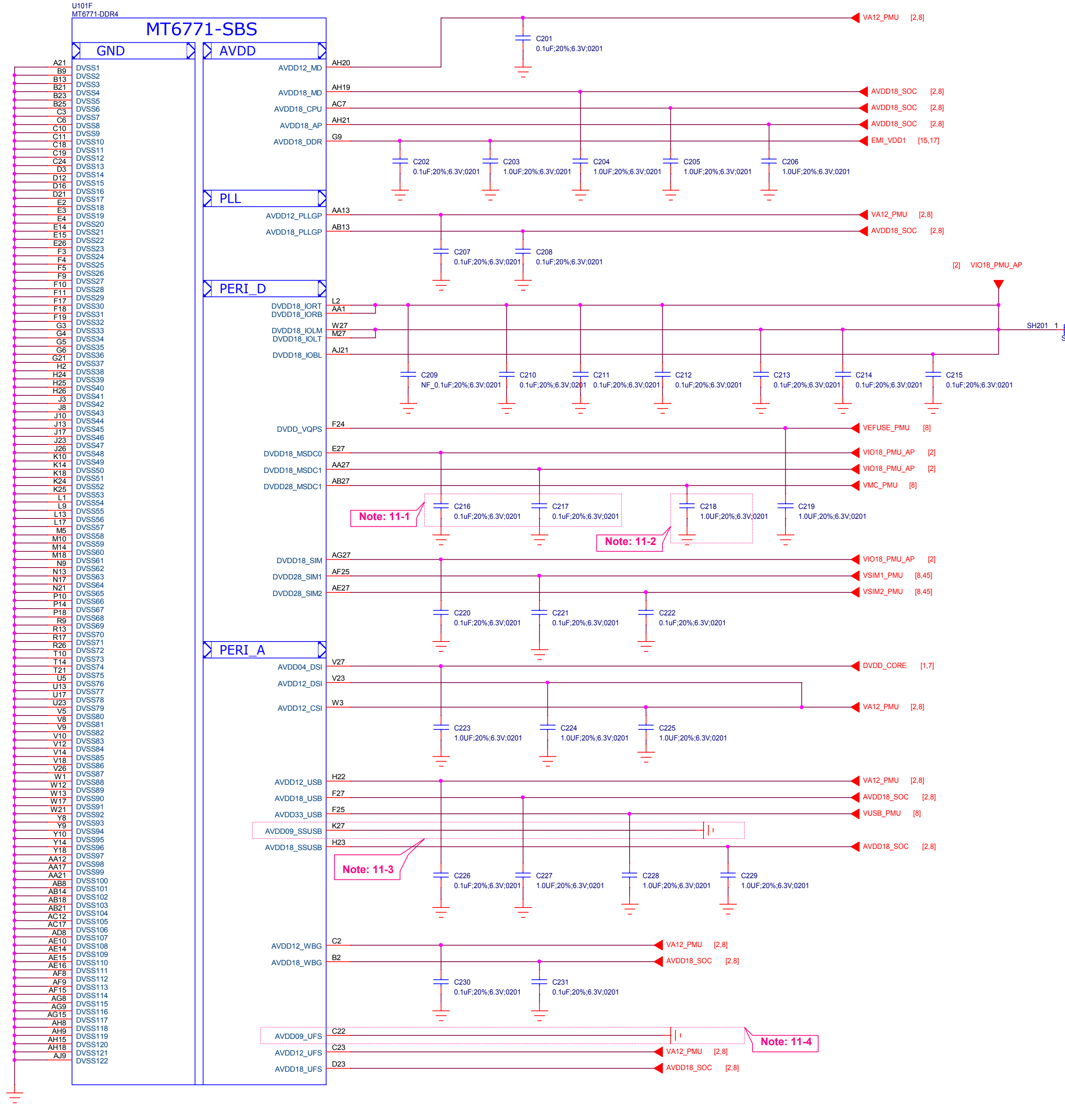
Note 10-1: Differential pair of DVDD_GPU remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-2: Differential pair of DVDD_PROC remote sense must be close to BB's ball
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-3: Differential pair of DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-4: Differential pair of DVDD_CORE remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Title	01_BB_POWER_PDN	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Thursday, May 30, 2019	Sheet 3	of 65



Schematic design notice of "11_BB_POWER_IO" page.

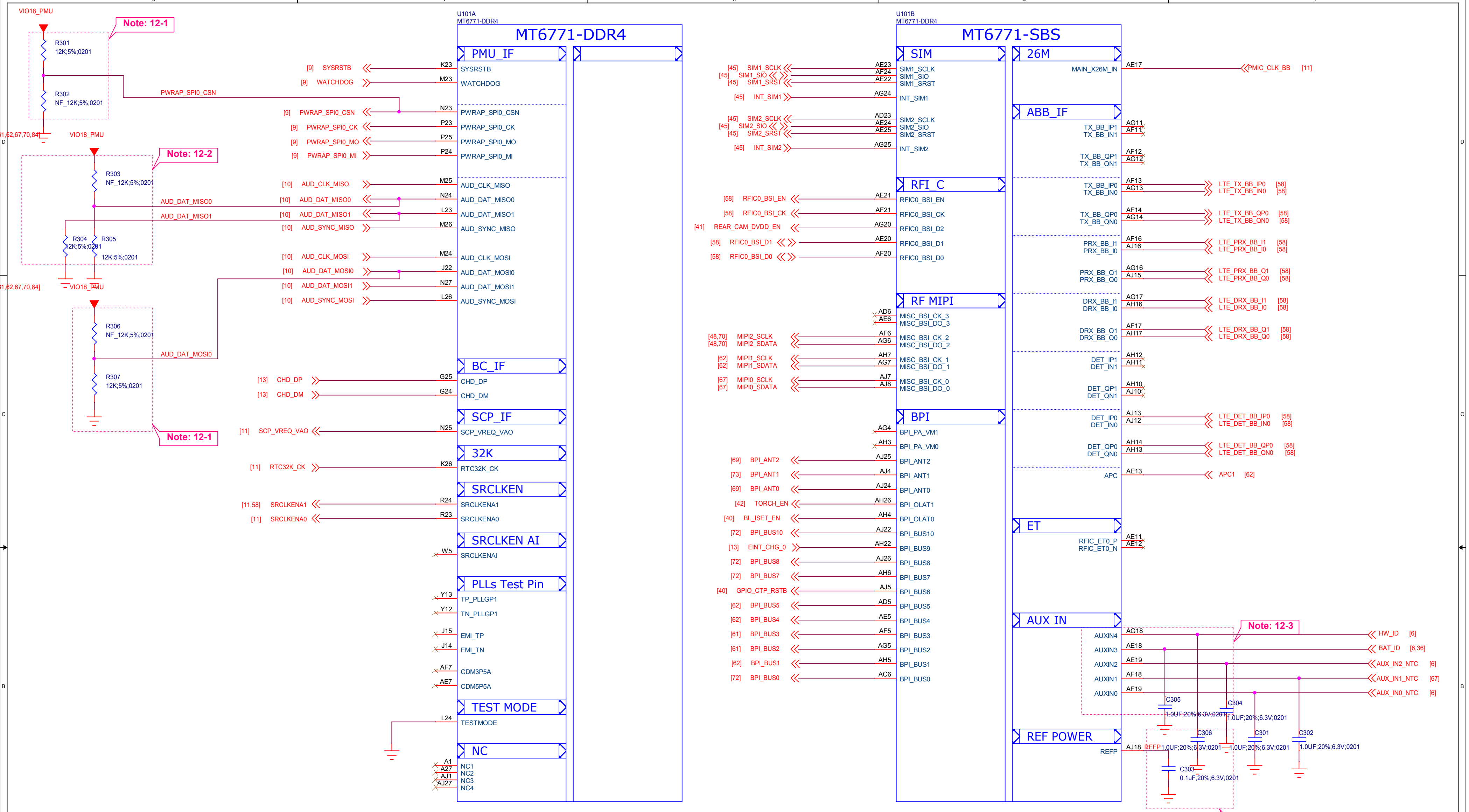
Note 11-1: C216 closed DVDD18_MSDC0 150mil
C217 closed DVDD18_MSDC1 150mil

Note 11-2: C218 closed DVDD28_MSDC1 150mil

Note 11-3: Connects "AVDD09_SSUSB" to GND when USB3.0 is not used.

Note 11-4: Connects "AVDD09_UFS" to GND when UFS is not used.

Title	02_BB_POWER_IO	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 4	of 65



Schematic design notice of "12_BB_1" page.

Note 12-1: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP_JTAG	IO_JTAG
HI	LO	N/A	N/A
HI	HI	SPI_CSB/SPI_CLK/ SPI_MO/SPI_MI/EINT8	N/A
LO	LO	SPI_CSB/SPI_CLK/ SPI_MO/SPI_MI/EINT8	DPI_11/DPI_HSYNC/DPI_VSYNC/DPI_DE/ DPI_CK/DPI_D8/DPI_D9
LO	HI	MSDC1_CLK/CMD/ DAT0/DAT1/DAT2	N/A

Note 12-2: "AUD_DAT_MISO0" is bootstrap pin to enable serial JTAG output over USB2.0 interface or not.
When "AUD_DAT_MISO0" is pulled to high in system start up and then USB2.0 interface will be switched into serial JTAG mode.

"AUD_DAT_MISO1" is bootstrap pin to select system booting up from eMMC or UFS device.

AUD_DAT_MISO1	Booting device
LO	eMMC
HI	UFS

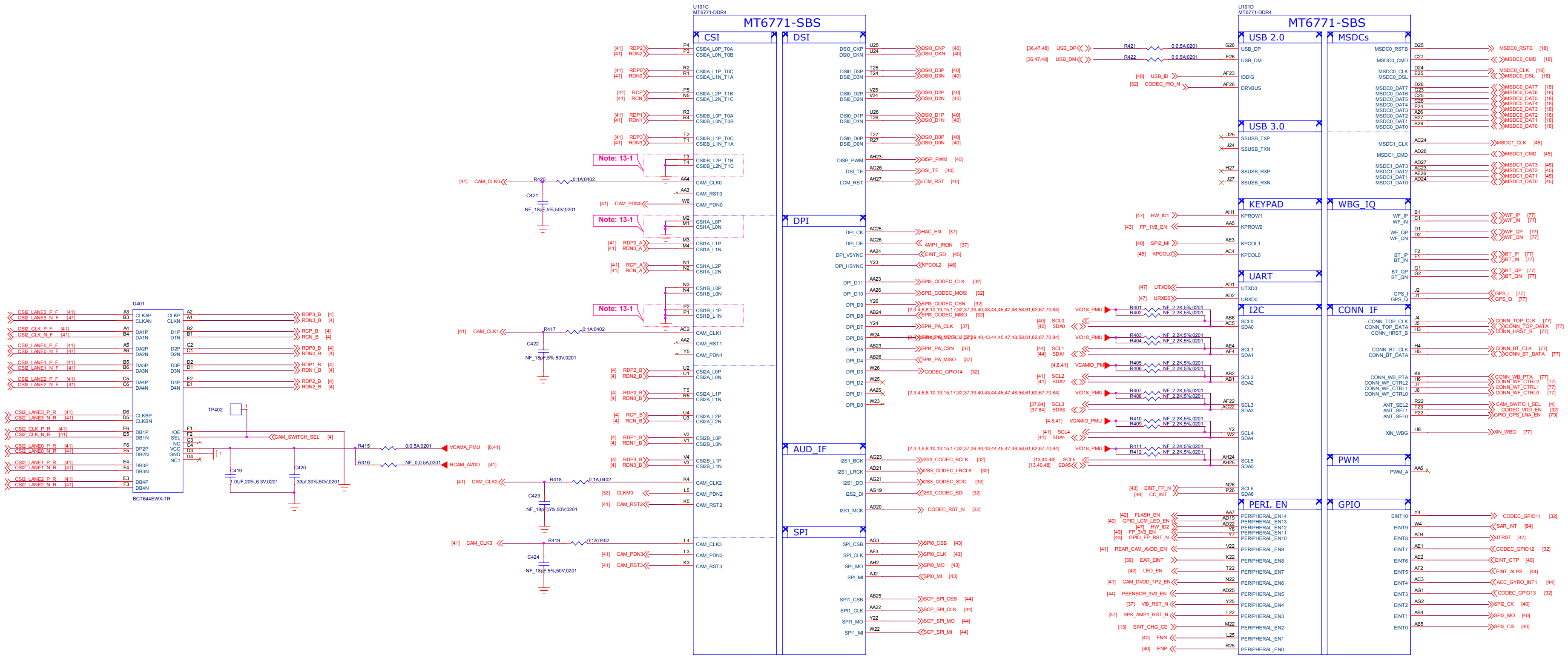
Note 12-3: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-4: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

Note 12-5: AUD_SYNC_MISO and AUD_CLK_MISO are DDR type feature in bootstrap

AUD_SYNC_MISO	AUD_CLK_MISO	DDR
LO	LO	LPDDR4X
LO	HI	LPDDR4X(Ext x 2 EN)
HI	LO	LPDDR3
HI	HI	LPDDR4X(Ext x 1 EN)

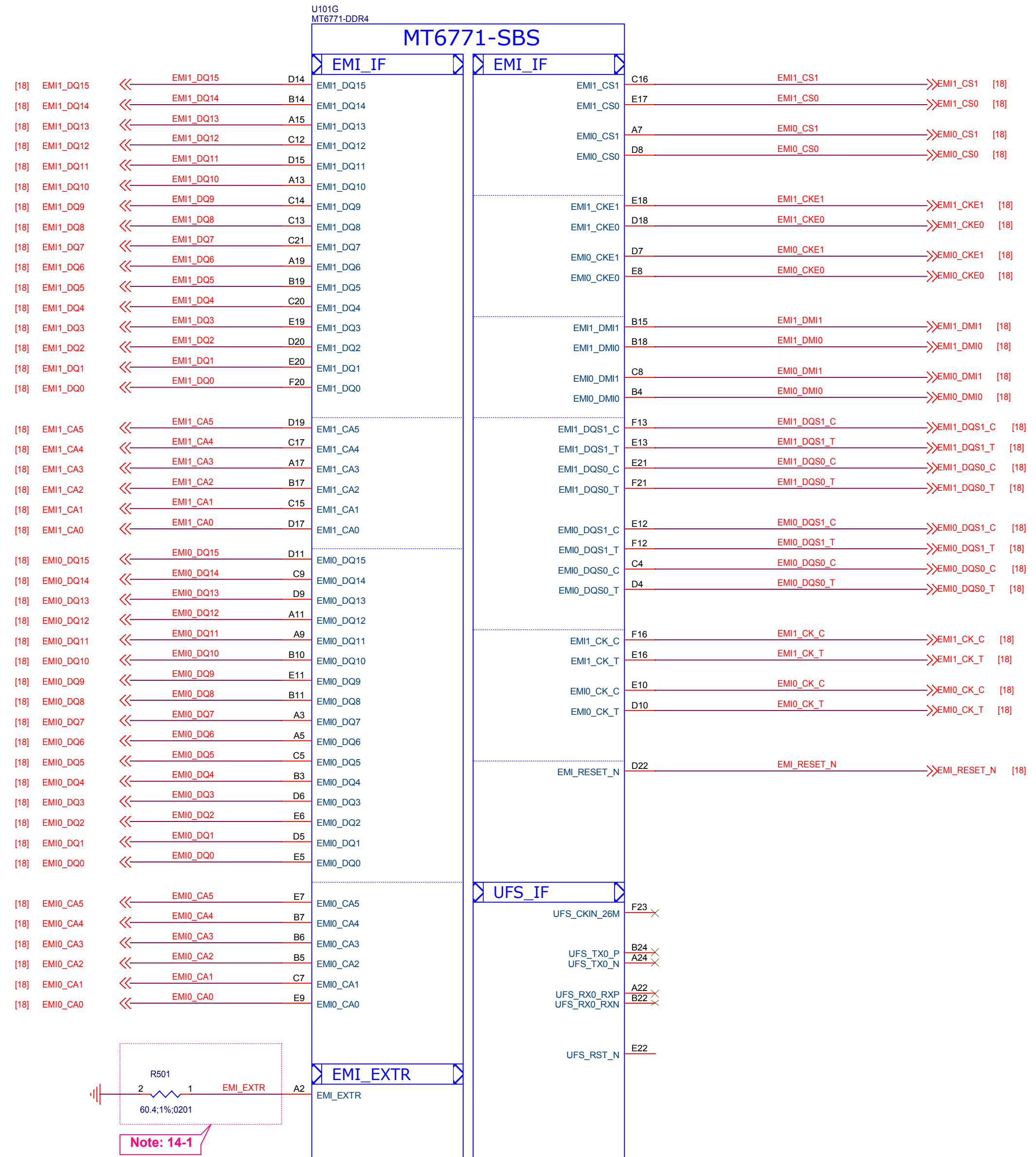
Title	03_BB_1_RF&SIM_IF	REV: V10	
DOCUMENT NO.:	Design Name	Size	C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet	5 of 65



Schematic design notice of "13_BB_2" page.

Note 13-1: CSI ports which are no use could be connected to GND or set in NC.
For detail information, please refer to MT6771 Design Notice

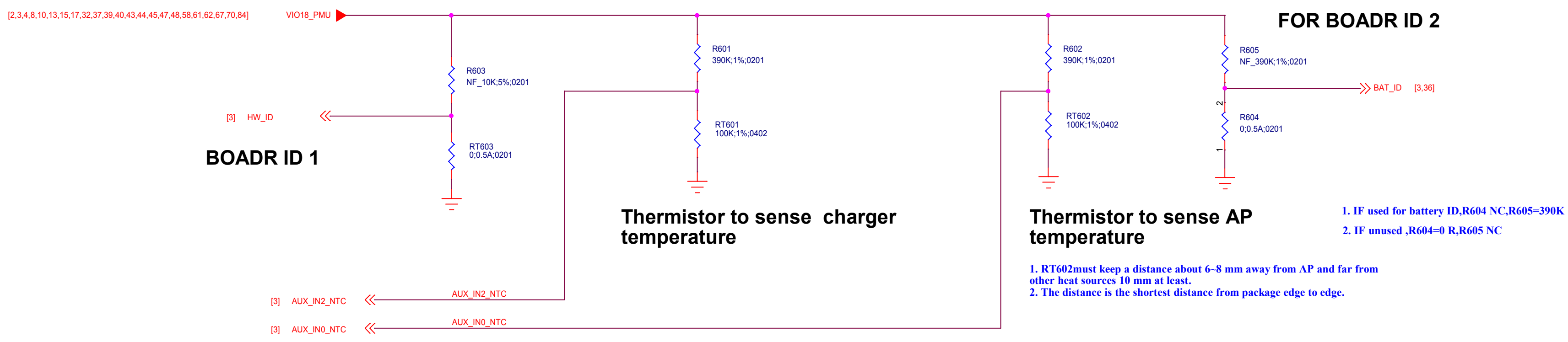
Title	04_BB_2_MIPI&GPIO	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 6 of 65	



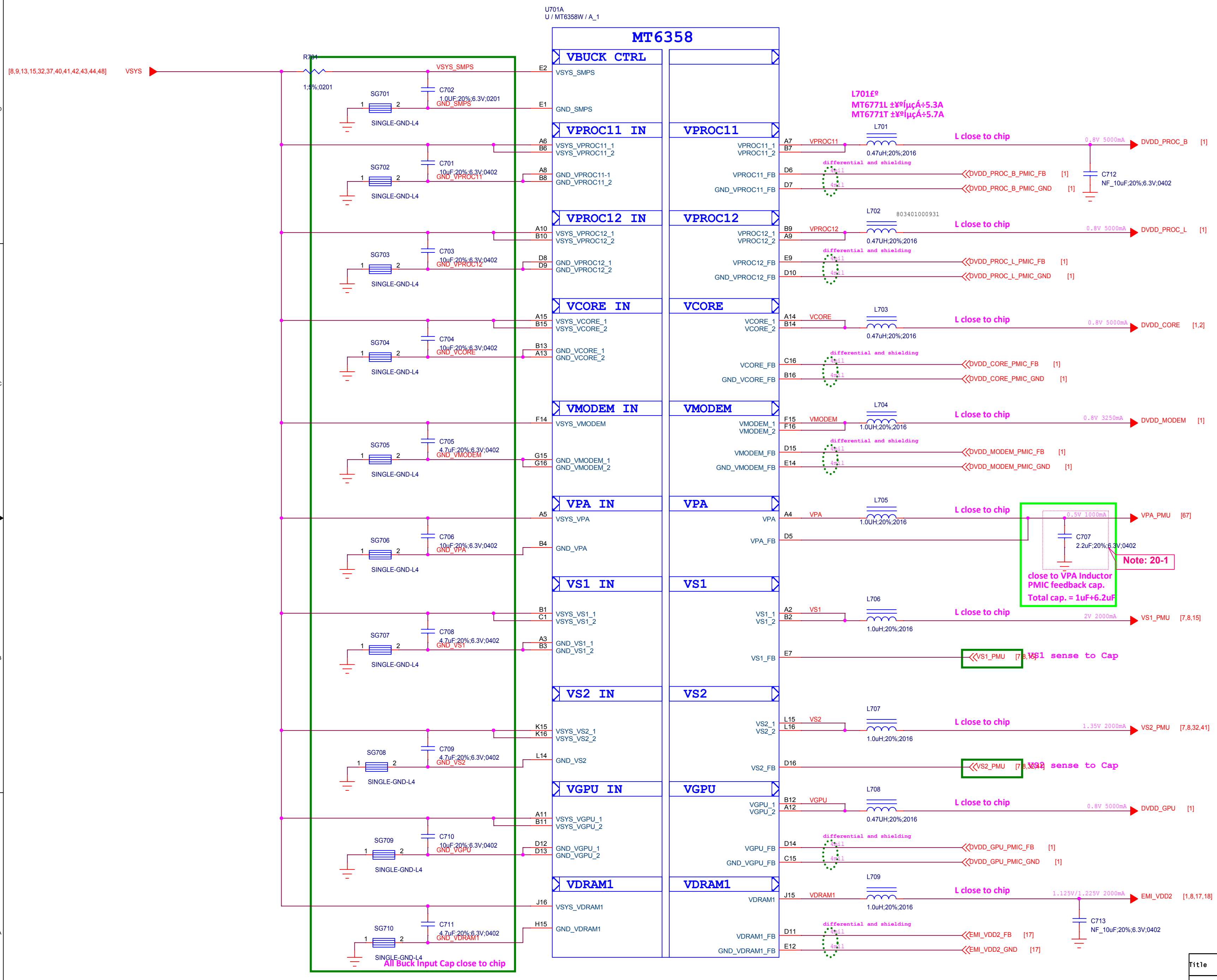
Schematic design notice of "14_BB_3" page.

Note 14-1: The resistor of EMI_EXTR for DRAM has to be placed near to BB as close as possible
R501, please select 60.4 ohm 1% resistor

Title	10_BB_ POWER_PDN	REV: V10
DOCUMENT NO.:	Design Name	Size C
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
Date: Thursday, May 30, 2019	Sheet 7 of 65	



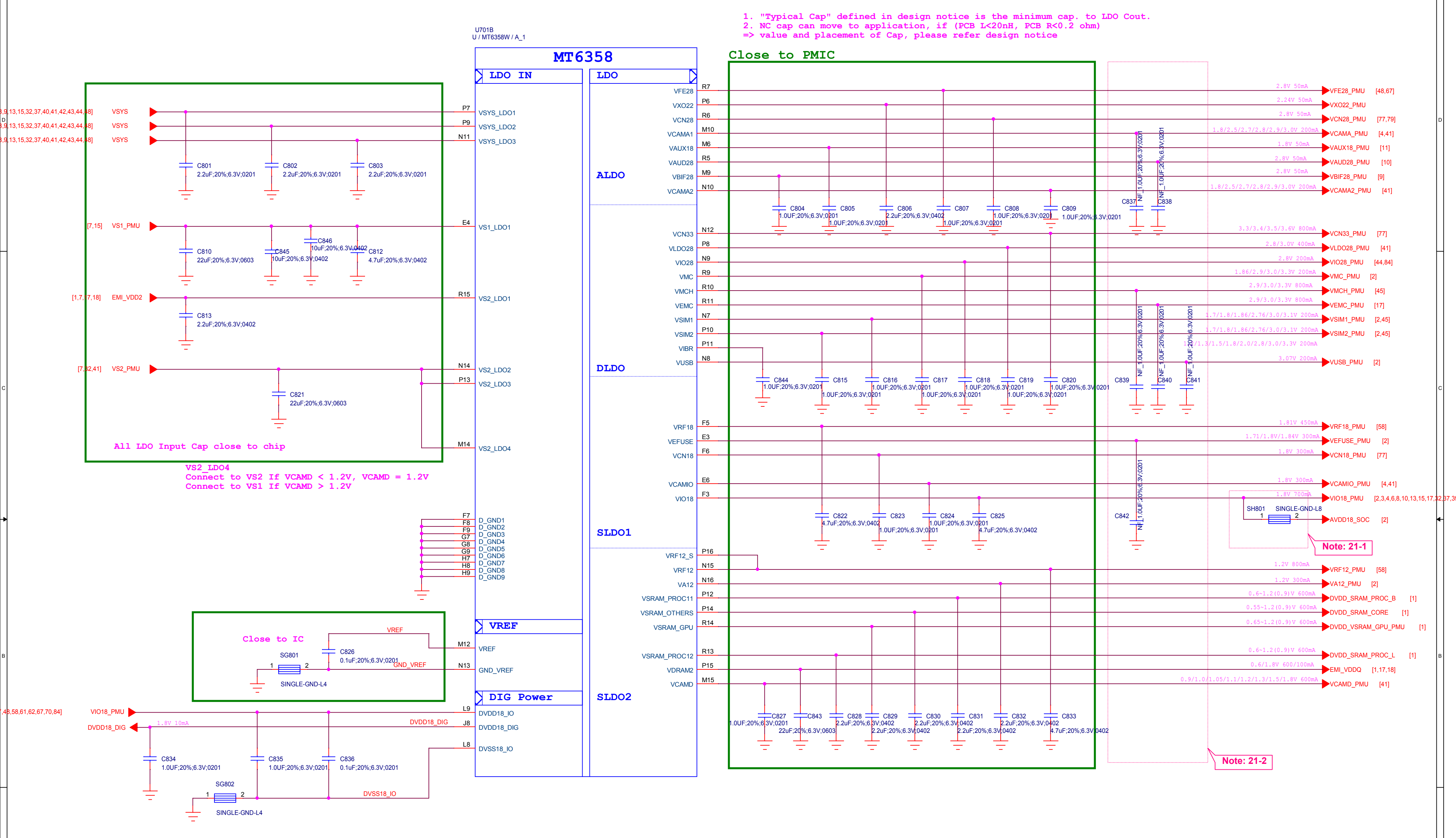
Title	06_BB_AUXADC_Thermal	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 8	of 65



Schematic design notice of "20_POWER_MT6358-Buck" page.

Note 20-1: Please select C707 with 0402 size

Title	07_POWER_MT6358-Buck	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Thursday, May 30, 2019	Sheet 9	of 65

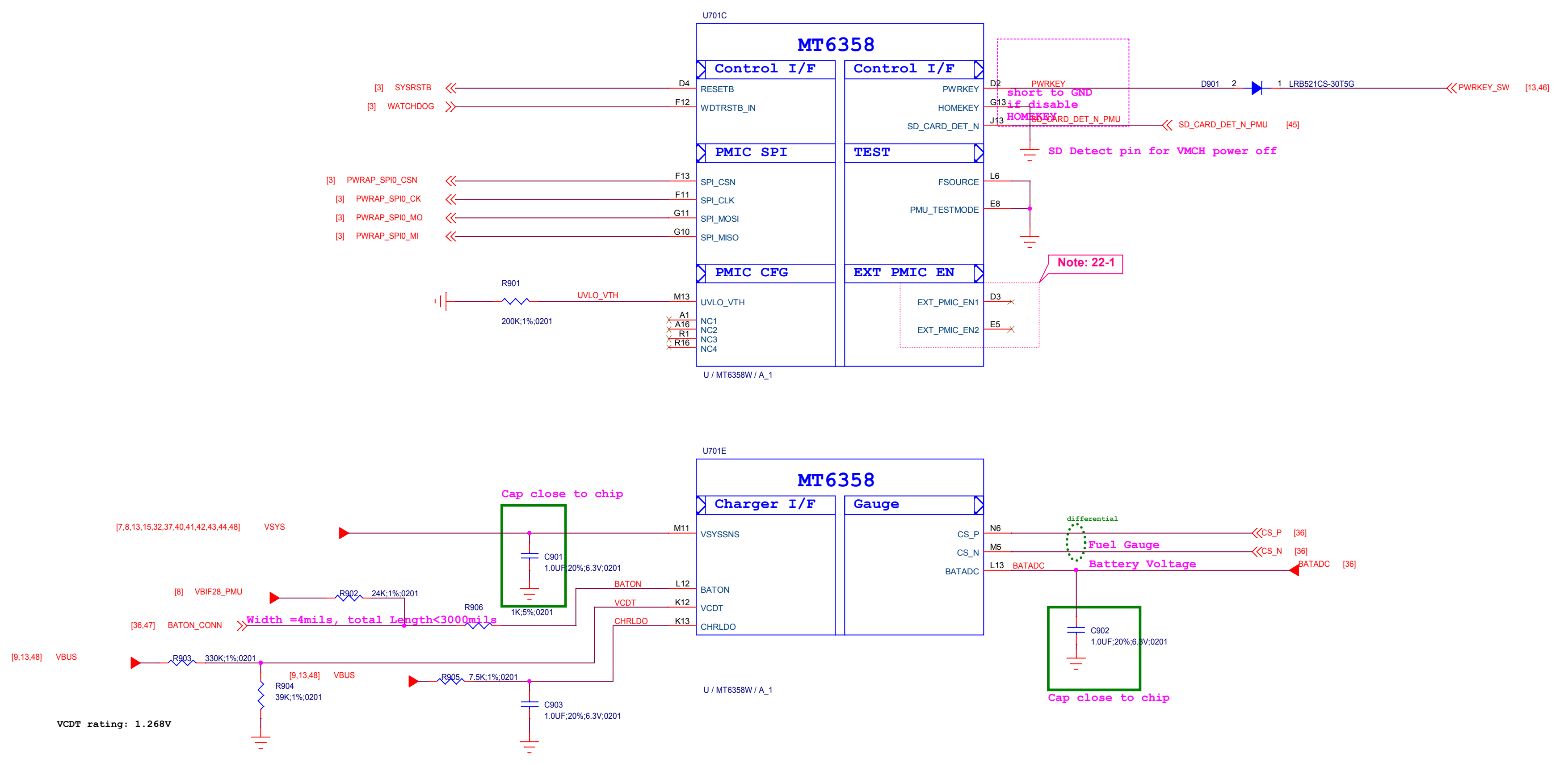


Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set SH801 close to C825, making star connection between VIO18_PMU and AVDD18_SOC near to LDO cap. C825
Please also refer to MT6358 design notice for further detail design information

Note 21-2: If these power trace can meet LDO layout constraint, these CAP can be NC or removed.
Please refer to MT6358 design notice.

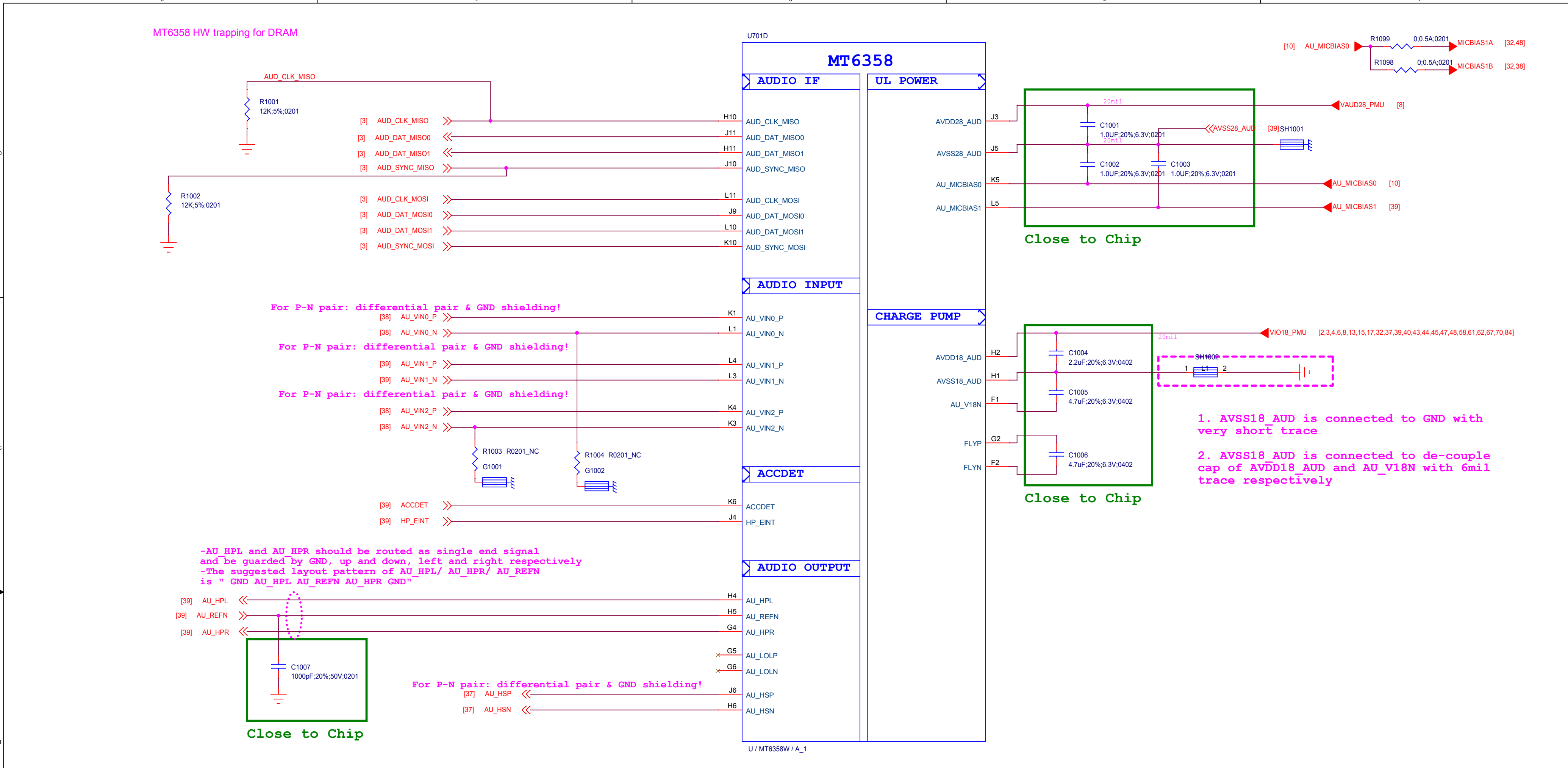
Title	08_POWER_MT6358-LDO	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI	
WINGTECH			
Date: Thursday, May 30, 2019		Sheet 10 of 65	



Schematic design notice of "22_POWER_MT6358-General"

Note 22-1: EXT_PMIC_EN1 : For UFS_1V8, and keep floating if it is not used
EXT_PMIC_EN2 : For VA09 of SSUSB/UFS, and keep floating if it is not used

Title	09_POWER_MT6358-General	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 11 of 65	



Schematic design notice of "23_POWER_MT6358-Audio"

Note 23-1: VDRAM 2 / VDRAM1 output voltage vs. trap pin.

HW GPIO configuration		Trapping Option		DRAM type	VDRAM2 Power source (VS2_LDO1_ball)
AUD_SYNC_MISO	AUD_CLK_MISO	VDRAM1	VDRAM2		
0	0	1.125V	0.6V	LP4X	VDRAM1
0	1	OFF	1.8V	LP4X (Ext x 2 EN)	VS1
1	0	1.225V	OFF	LP3	VDRAM1
1	1	1.125V	1.8V	LP4X (Ext x 1 EN)	VS1

Title	10_POWER_MT6358-Audio
REV:	V10
DOCUMENT NO.:	Design Name
Size	C
DEPARTMENT:	WINGTECH-SH
DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019
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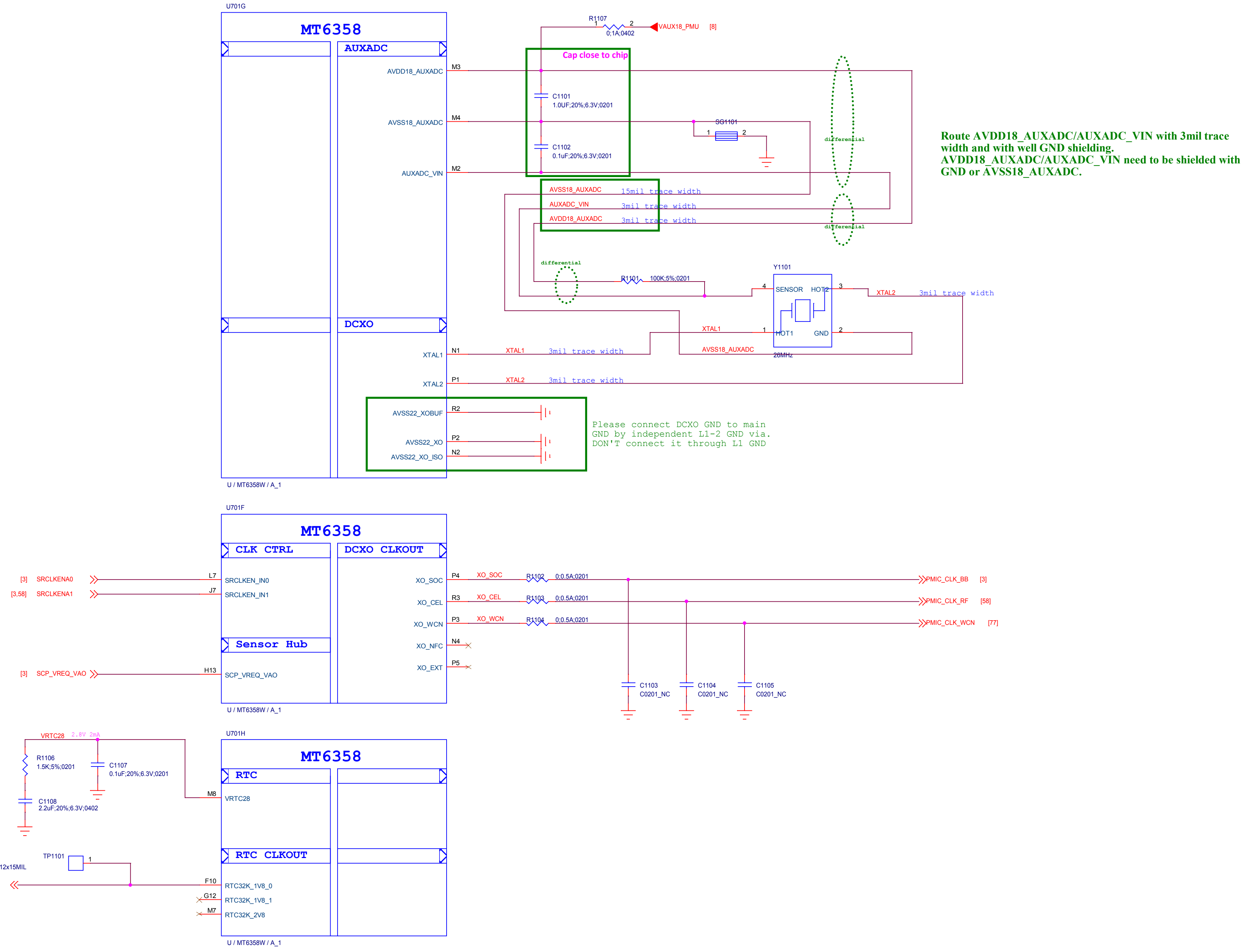
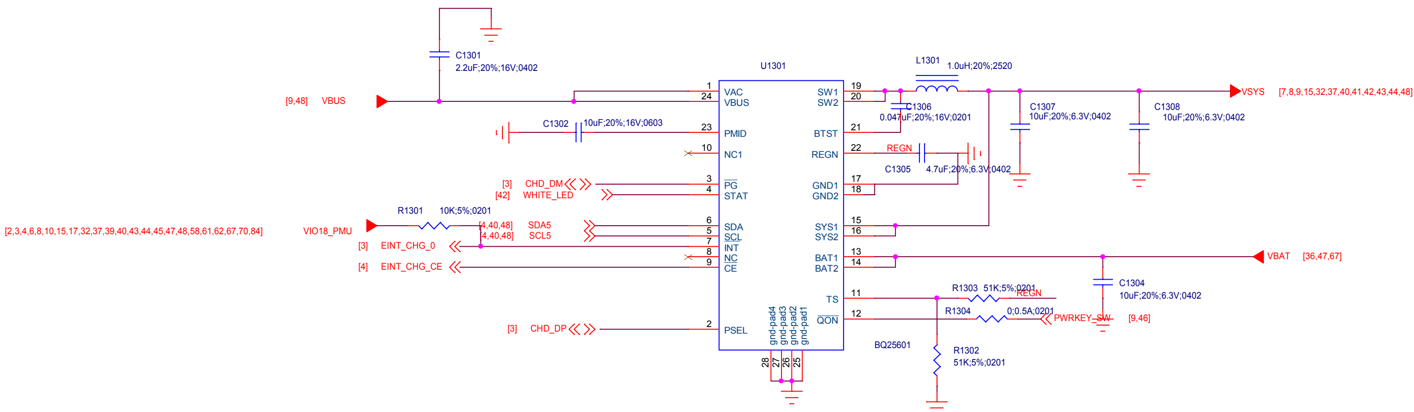


Title	11_POWER_MT6358_Clock		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 13	of 65

Title	12_POWER_MT6370-General		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 14 of 65	



Title 13_POWER_MT6370-Charger + PP		REV: V10
DOCUMENT NO.: Design Name		Size C
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
Date: Thursday, May 30, 2019	Sheet 15 of 65	

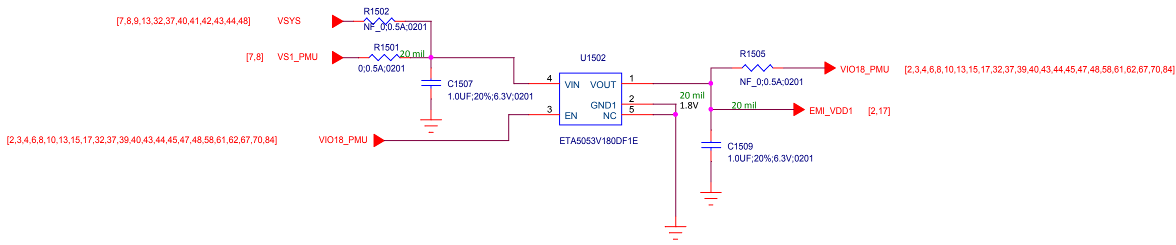
Schematic design notice of "27_POWER_SubPMIC-HV powers" page.

Note 27-1: It is recommended to reserve 0-ohm and cap. for BOM fine tune to minimize RF de-sense.

Note 27-2: It is recommended to reserve 0-ohm for BOM fine tune to minimize RF de-sense.

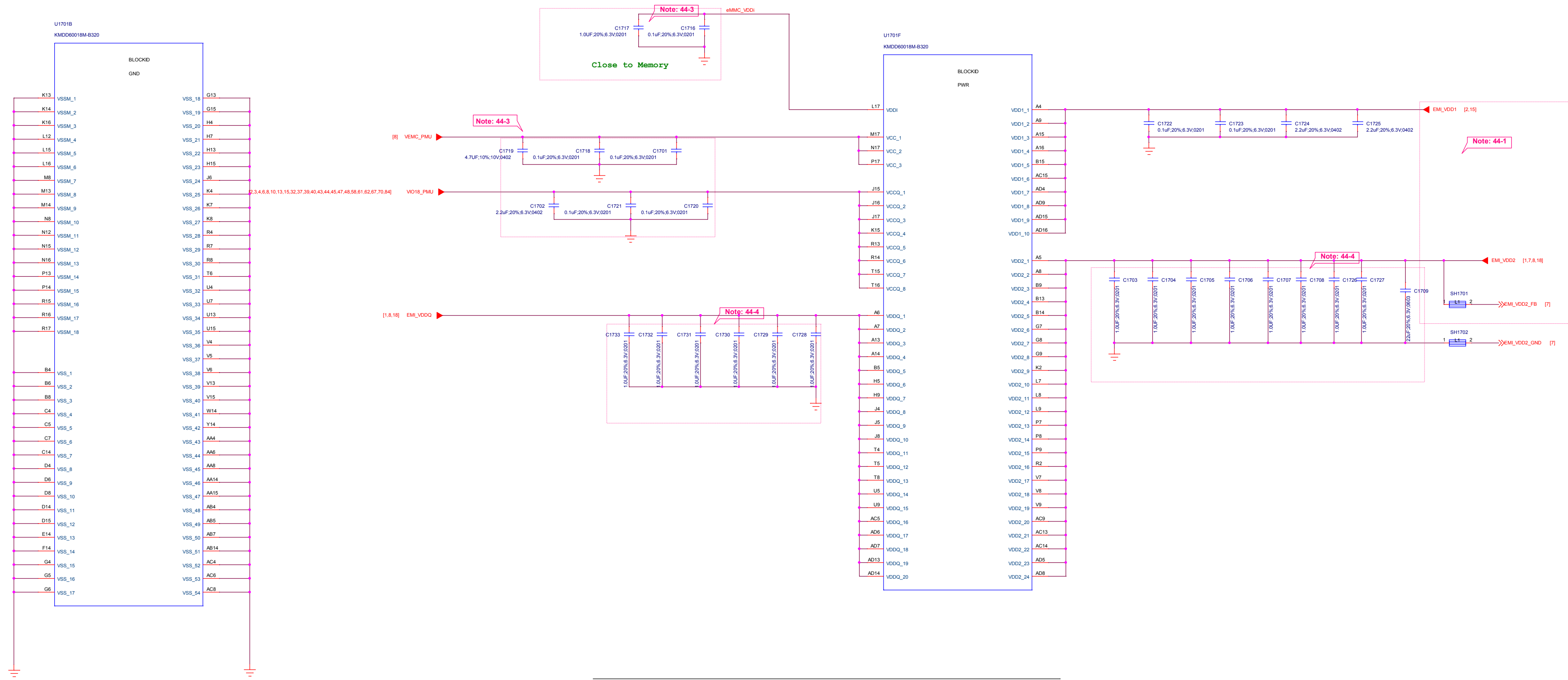
Title	14_POWER_MT6370-HV powers		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Thursday, May 30, 2019	Sheet 16 of 65	

LDO for EMI_VDD1 of LPDDR4 VDD1



Title	15_POWER_ThirdParty_Powers	REV: V10
DOCUMENT NO.:	Design Name	Size C
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
Date: Thursday, May 30, 2019	Sheet 17 of 65	

Title	16_BB_ POWER_PDN		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 18 of 65	



Schematic design notice of "44_Memory_eMMC_LPDDR4X"

Note 44-1:	Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X
Note 44-2:	DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,
Note 44-3:	Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.
Note 44-4:	VDD2 VDDQ decoupling cap: closed to DRAM ball. For other cap for PMIC [>10uF, at PMIC page]: please also refer to MMD and layout guide for placement.

Title	17_Memory_LPDDR4_POWER	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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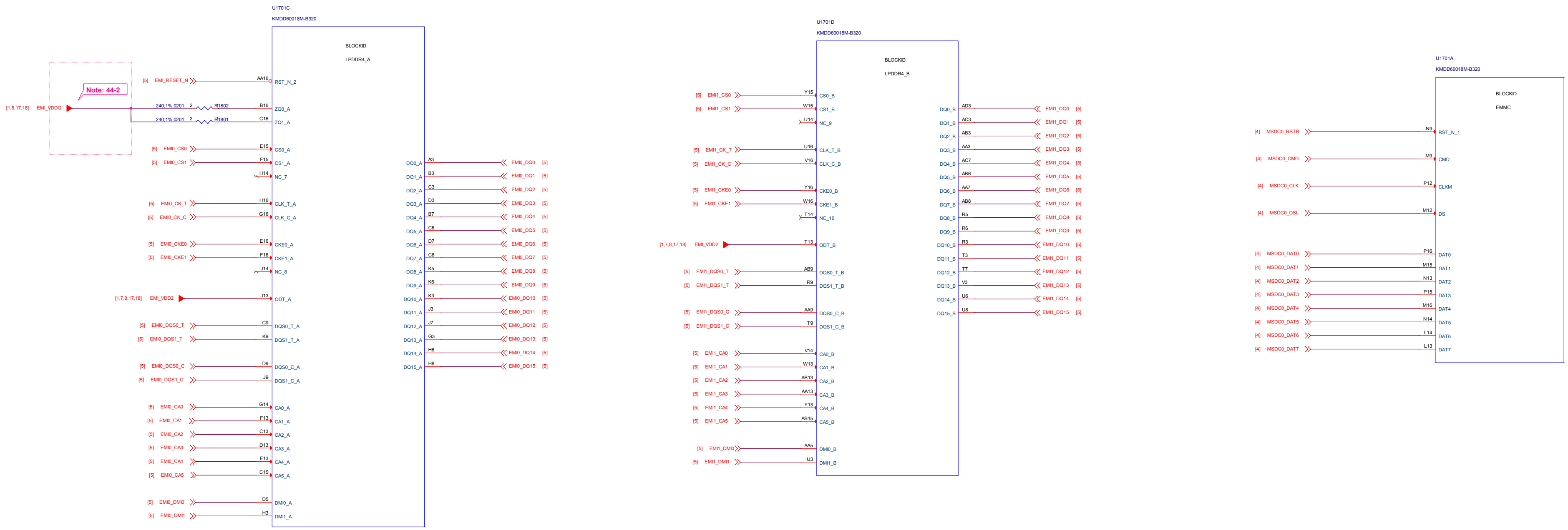
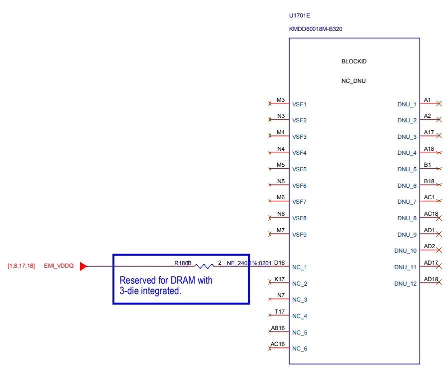




Schematic design notice of "44_Memory_eMMC_LPDDR4X"

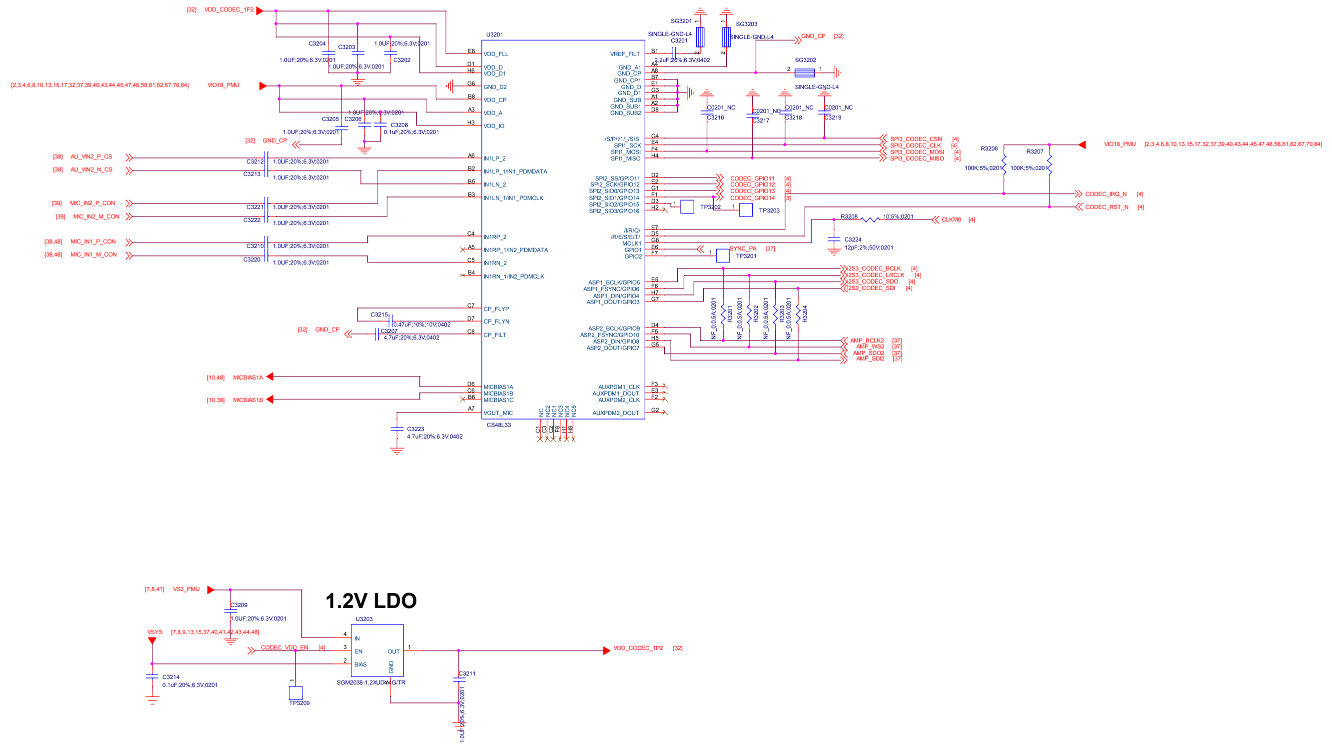
Note 44-1: Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

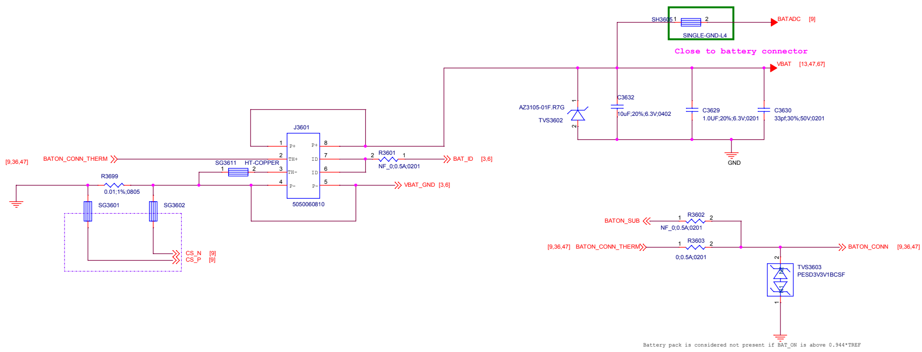
Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC [>10uF, at PMIC page]:
please also refer to MMD and layout guide for placement.

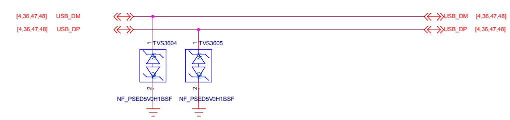
Title	18_Memory_eMMC_LPDDR4	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet	20 of 65



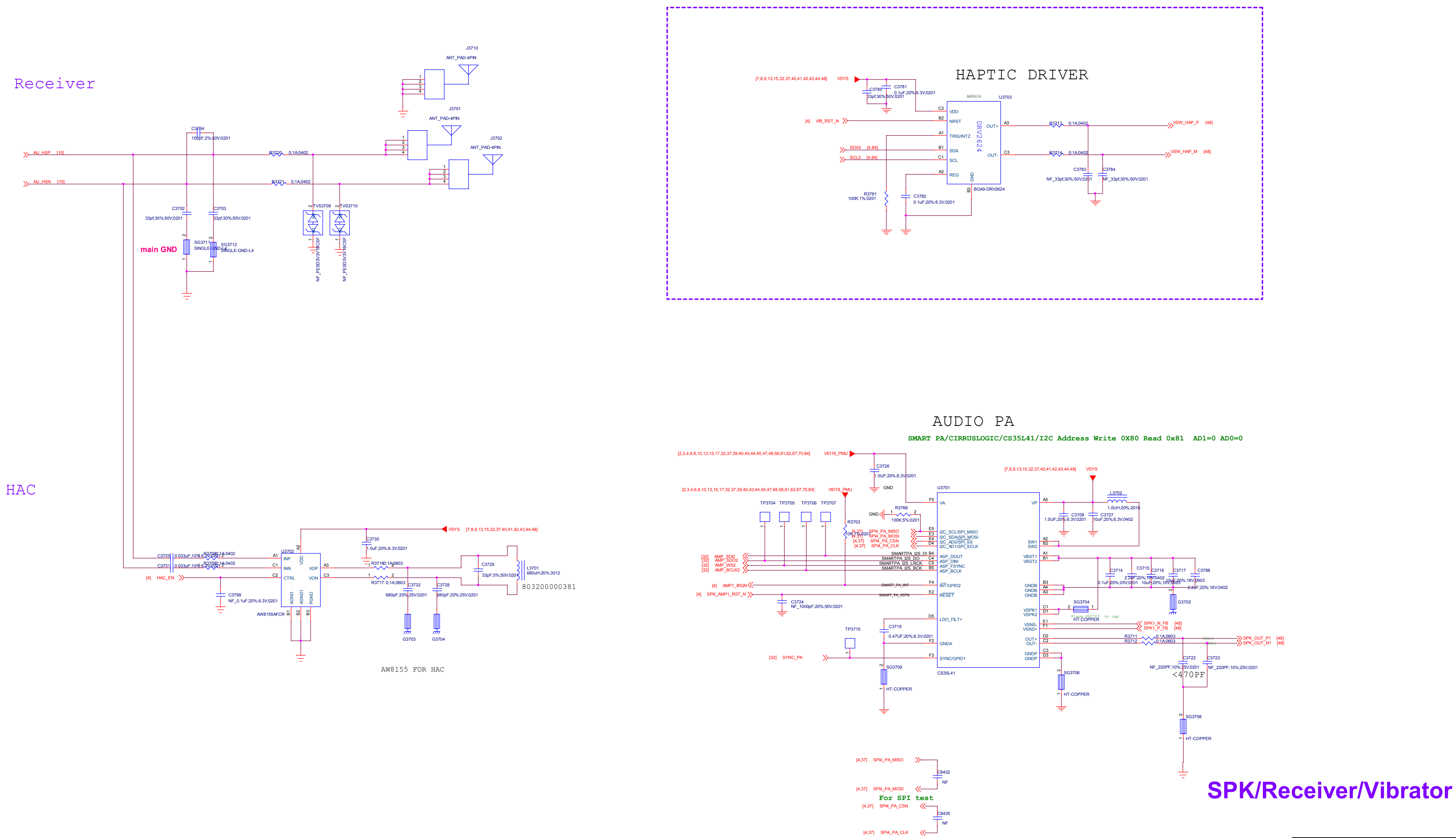
Title	32.CODEC_Cirrus		REV: V10
DOCUMENT NO.:	98860_1_12M13_201905302120		Size D
DEPARTMENT:	DEPARTMENT	DESIGNER:	DESIGNER
Date:	Thursday, May 30, 2019	Sheet 21 of 65	

BATTERY CONNECTOR



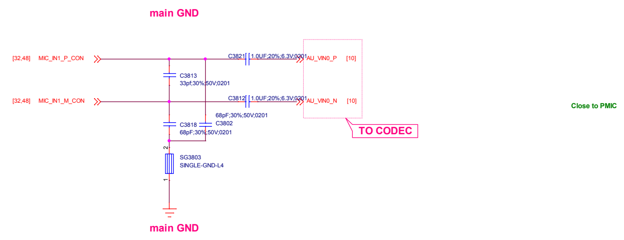


Title	36_Battery/USB IF	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 22 of 65

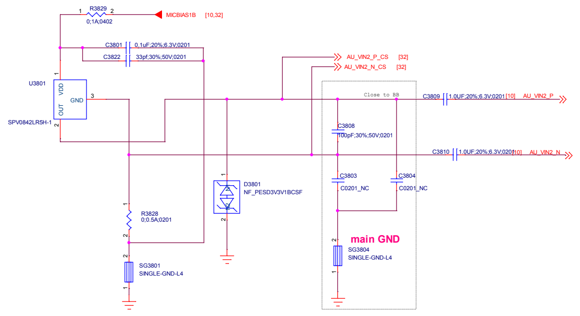


Title	37_SPK/Receiver/Vibrator	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet	23 of 65

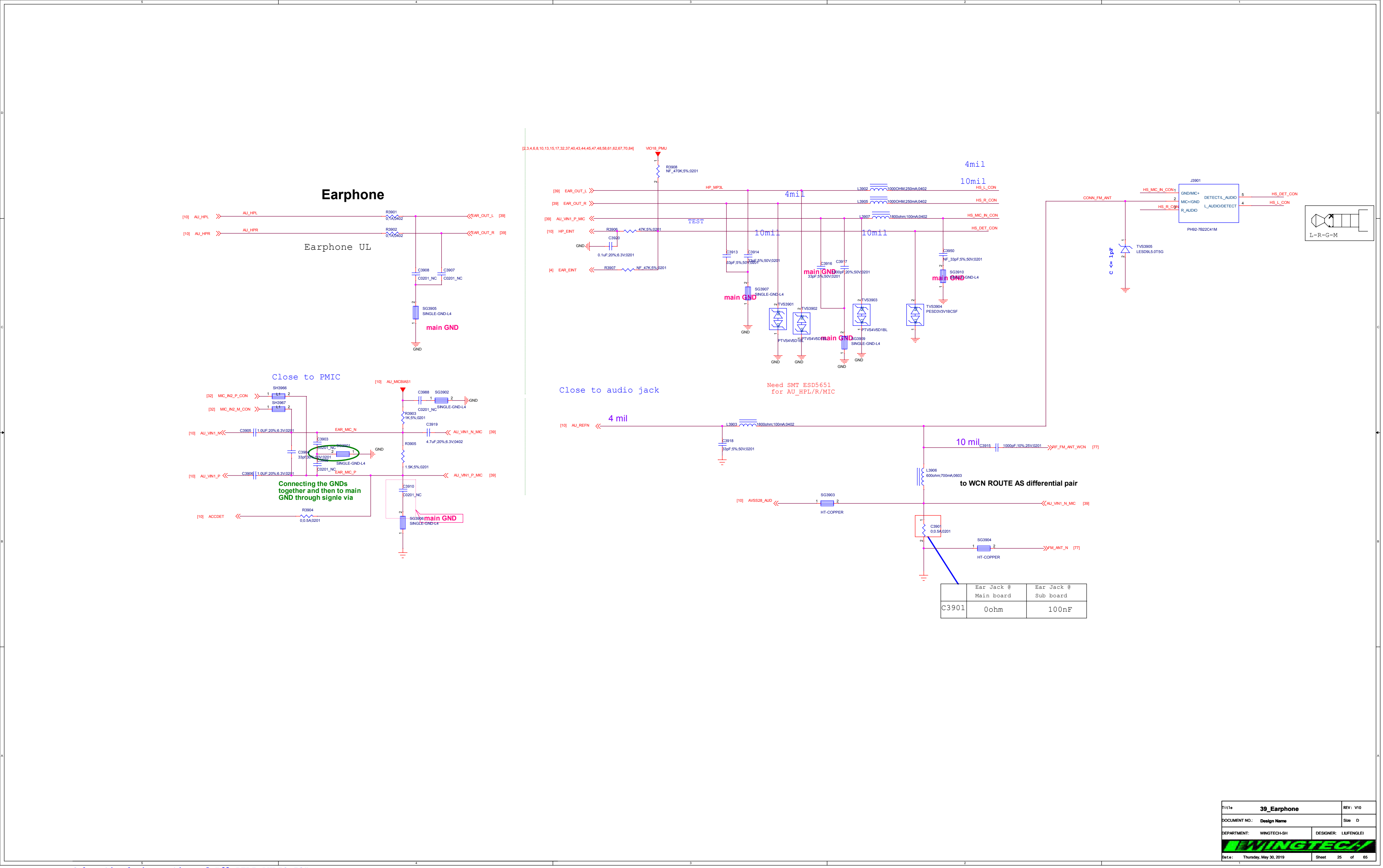
MAIN MIC



TOP MIC



Title	38_MIC	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 24 of 65	



Schematic design notice of "62_PERI_AUDIO_IO" page.

Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

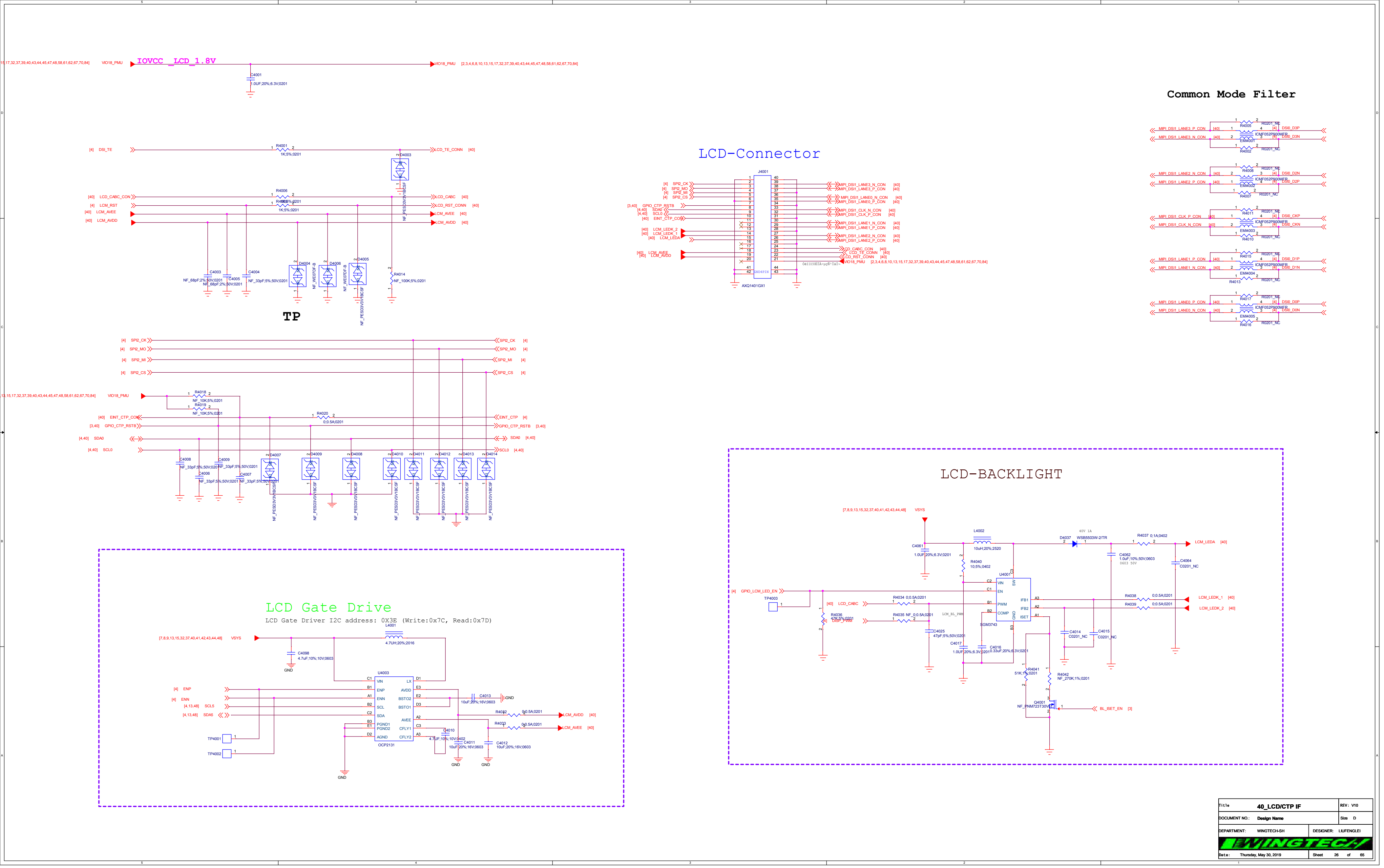
Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Note 62-3:

	Earphone Jack @ Main board	Earphone Jack @ Sub board
R6513	0 ohm	100nF

Note 62-4: Please Select ACC Mode for Operator Project to Pass Electrical MOS Test; More Information refer to Audio/Speech Design Notice

Note 62-5: Please select R6231 with 0402 size



IOVCC _LCD_1.8V
Common Mode Filter
LCD-Connector
TP
LCD-BACKLIGHT
LCD Gate Drive
LCD Gate Driver I2C address: 0X3E (Write:0x7C, Read:0x7D)
U4003
OCP2131
U4001
SGM3743
J4001
AXG1401GX1
Title 40_LCD/CTP IF
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
WINGTECH
Date: Thursday, May 30, 2019
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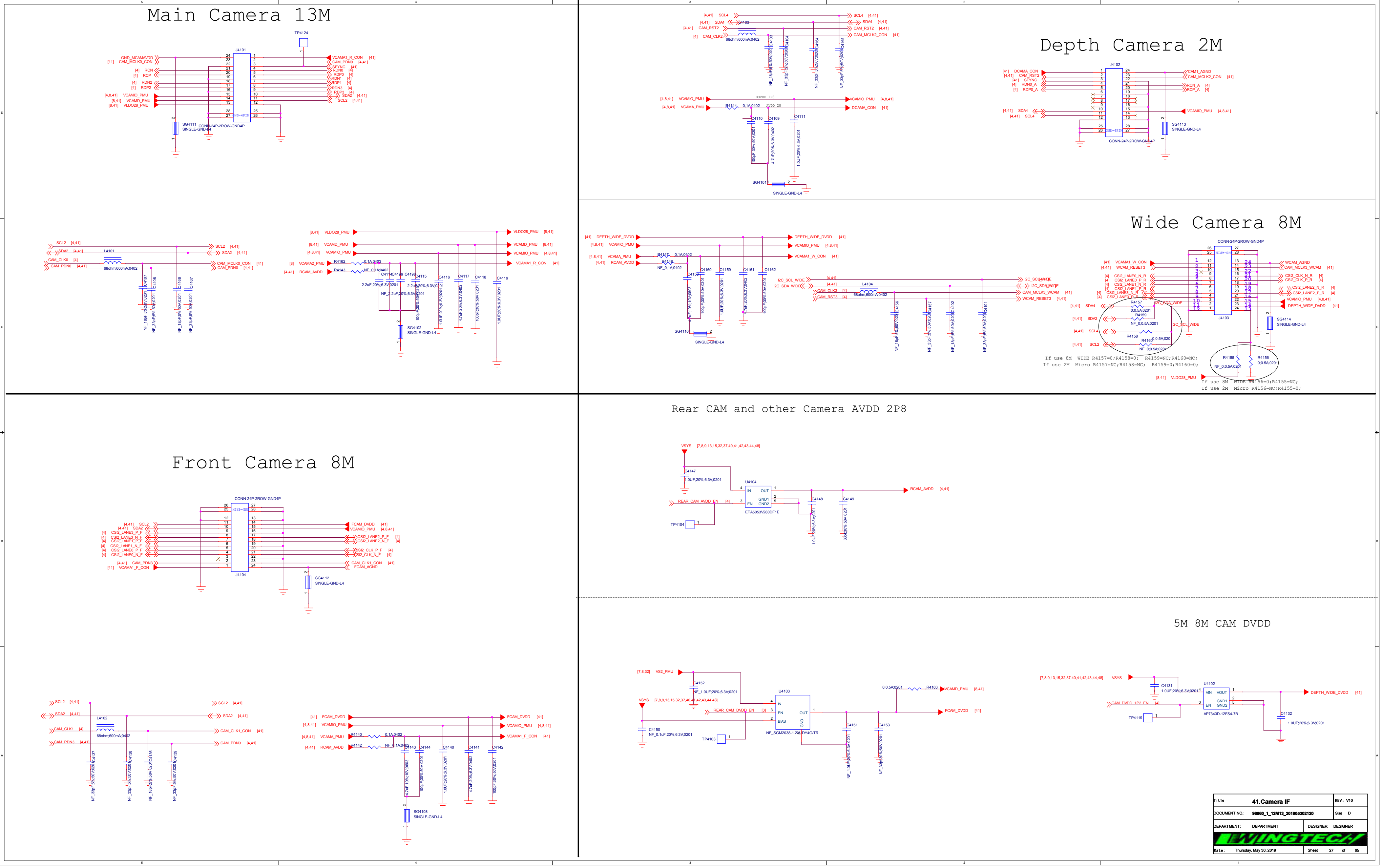
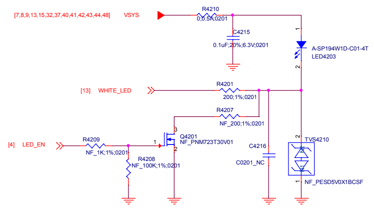
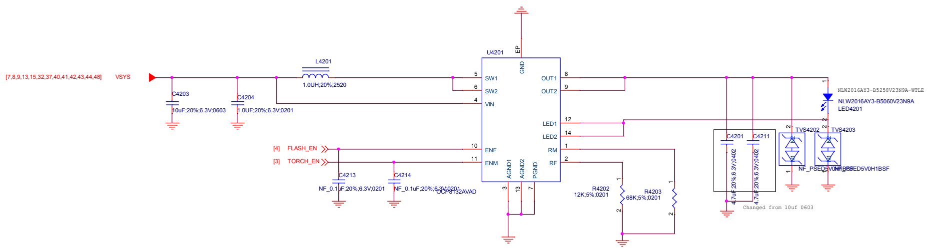

Main Camera 13M
Depth Camera 2M
Wide Camera 8M
Front Camera 8M
Rear CAM and other Camera AVDD 2P8
5M 8M CAM DVDD
If use 8M WIDE R4157=0;R4158=0; R4159=NC;R4160=NC;
If use 2M Micro R4157=NC;R4158=NC; R4159=0;R4160=0;
If use 8M WIDE R4156=0;R4155=NC;
If use 2M Micro R4156=NC;R4155=0;
Title 41.Camera IF
REV: V10
DOCUMENT NO.: 98860_1_12M13_20190530212O
Size D
DEPARTMENT: DEPARTMENT
DESIGNER: DESIGNER
WINGTECH
Date: Thursday, May 30, 2019
Sheet 27 of 65



Input and output cap voltage confirm ?



Title	42_Flash/RGB	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
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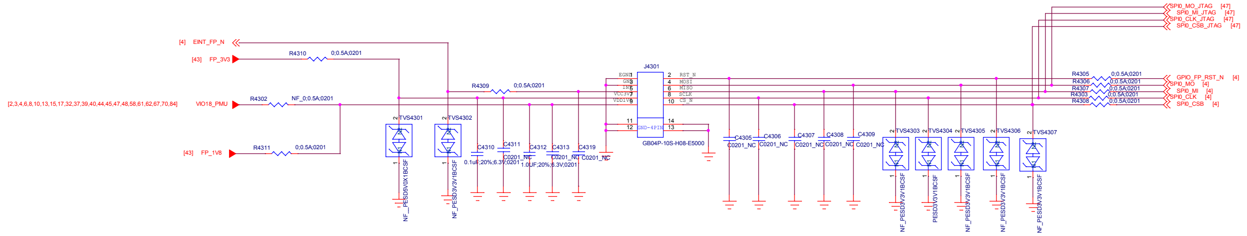
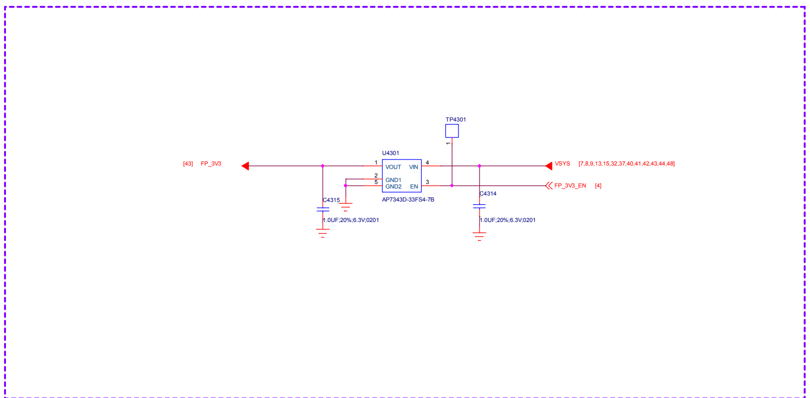
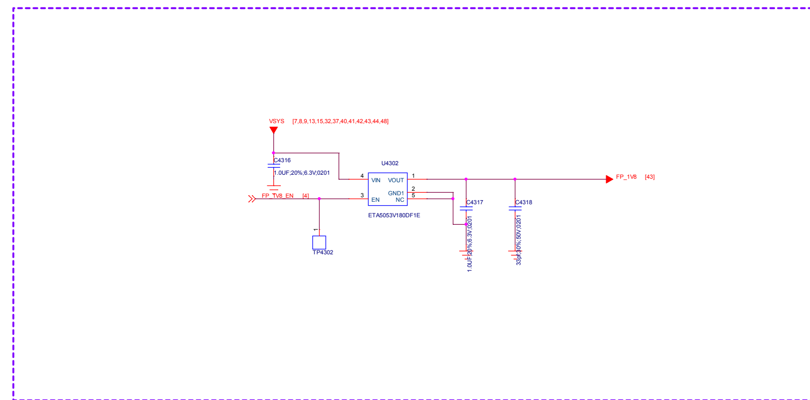
Fingerprint



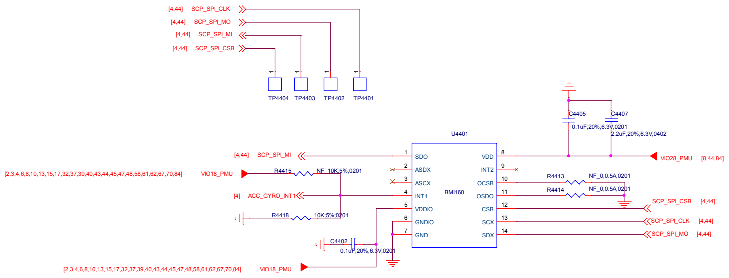
Fingerprint





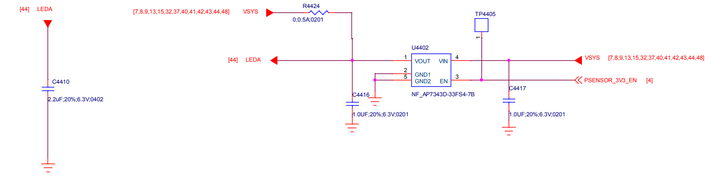
Title	43.Fingerprint		REV: V10
DOCUMENT NO.:	98860_1_12M13_201905302120		Size D
DEPARTMENT:	Wingtech-SZ	DESIGNER:	DESIGNER
Date:	Thursday, May 30, 2019	Sheet 29 of 65	

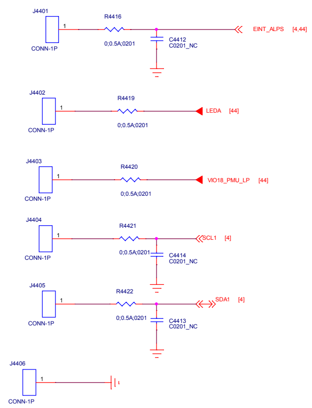
ACCELEROMETER+GYRO



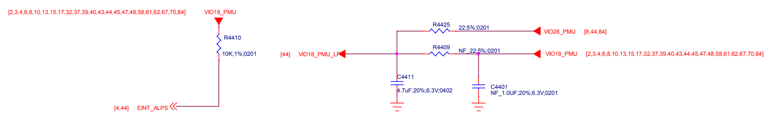
ALP-Sensor

Proximity Sensor

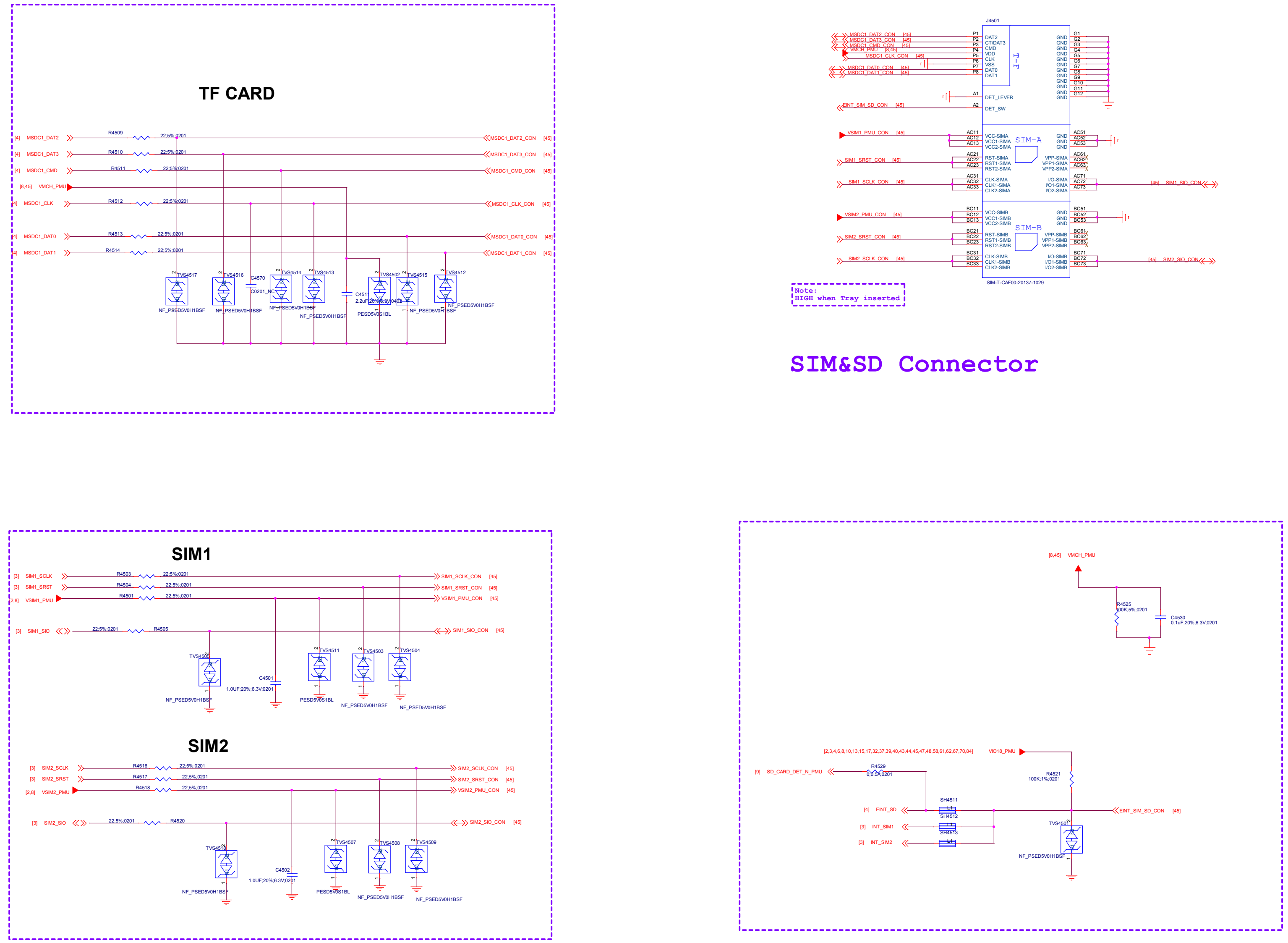




Ambient Light Sensor



Title	44_Sensor	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 30 of 65	



Title	45_SIM/TF IF	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT: WINGTECH-SH		DESIGNER:	LIUFENGLEI
WINGTECH			
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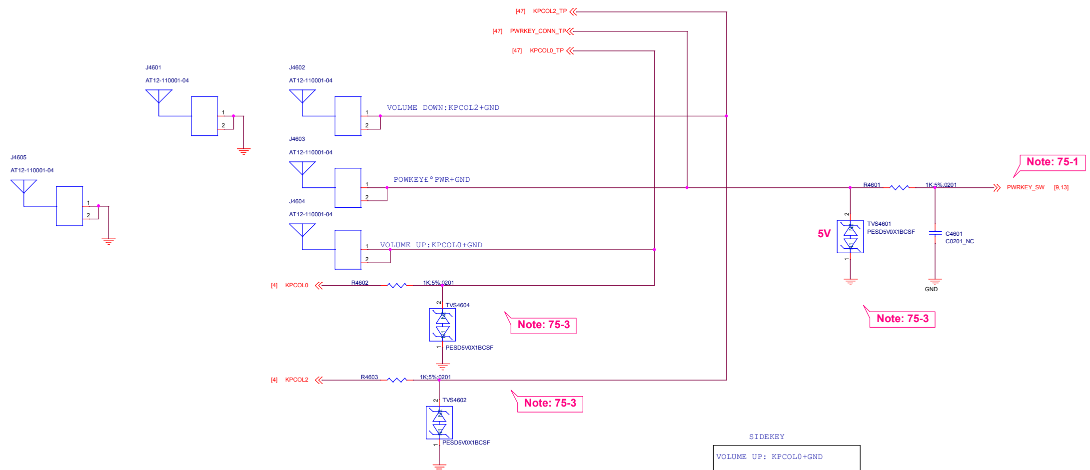
Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY

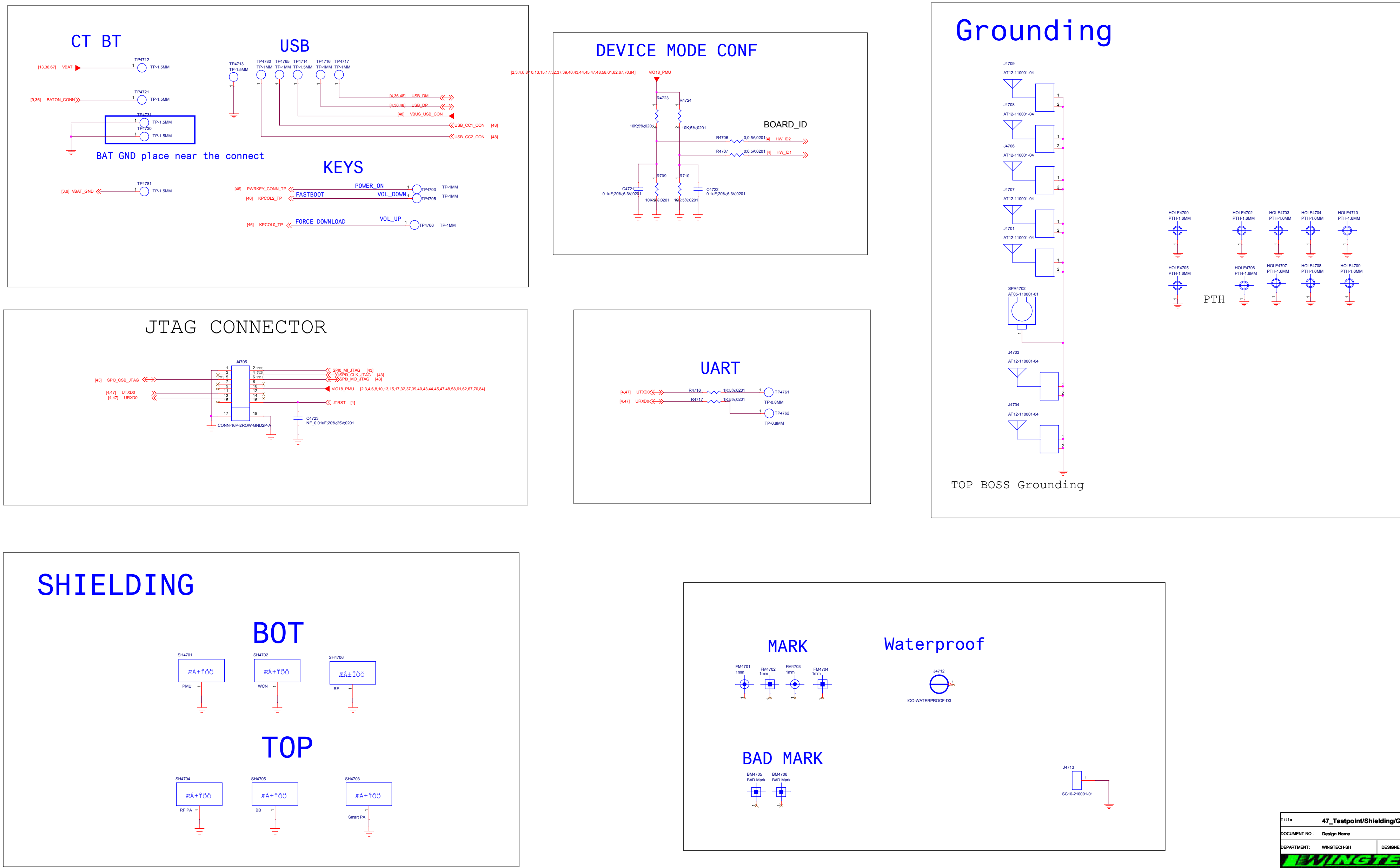


SIDEKEY

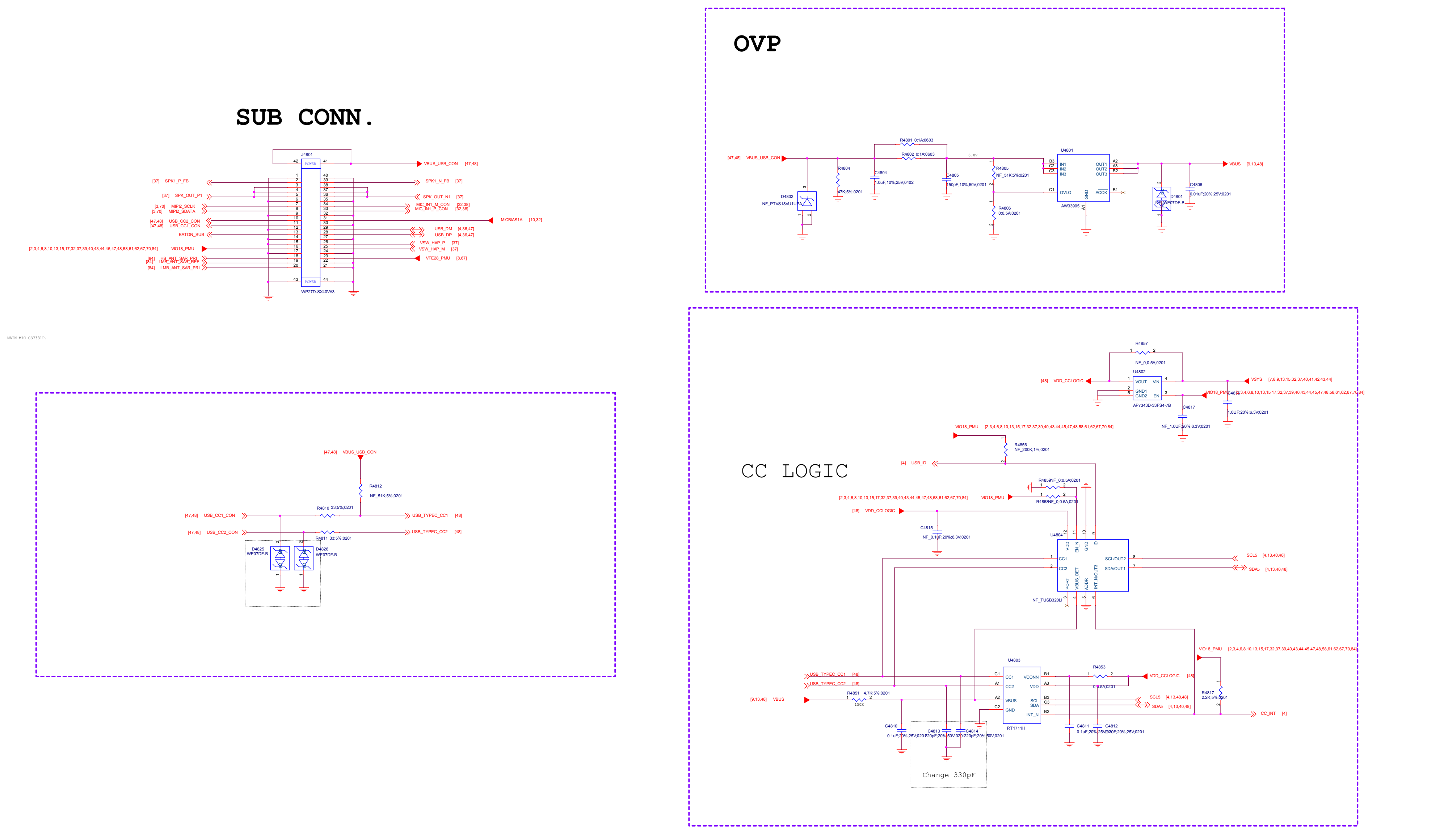
VOLUME UP: KPCOL0+GND
POWERON PWRKEY+GND
VOLUME DOWN: KPCOL2+GND

Note 75-1:DO NOT put pull-up resistor on PWRKEY
Note 75-2: Volume Up:KPCOL0+GND
Volume Down:KPCOL2+GND

Title	46_Sidekey	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 32 of 65	



Title	47_Testpoint/Shielding/GND	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 33	of 65



Title	48_Sub PCB IF		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 34	of 65

GPIO Interface

[3,69] BPI_ANT2
[3,73] BPI_ANT1
[3,69] BPI_ANT0

[3,72] BPI_BUS7

[3,62] BPI_BUS5
[3,62] BPI_BUS4
[3,61] BPI_BUS3
[3,61] BPI_BUS2
[3,62] BPI_BUS1
[3,72] BPI_BUS0

MIPI Interface

[3,67] MIPI0_SCLK
[3,67] MIPI0_SDATA

[3,62] MIPI1_SCLK
[3,62] MIPI1_SDATA

[3,48,70] MIPI2_SCLK
[3,48,70] MIPI2_SDATA

POWER Interface

[13,36,47,67] VBAT
[2,3,4,6,8,10,13,15,17,32,37,39,40,43,44,45,47,48,58,61,62,67,70,84] VIO18_PMU
[8,48,67] VFE28_PMU
[8,44,84] VIO28_PMU
[7,67] VPA_PMU
[8,77,79] VCN28_PMU
[8,77] VCN18_PMU
[8,77] VCN33_PMU

I2C Interface

[4,37,84] SDA3
[4,37,84] SCL3

Title	Page Name = 56_RF Interface	REV: V10	
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530	D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet 35	of 65

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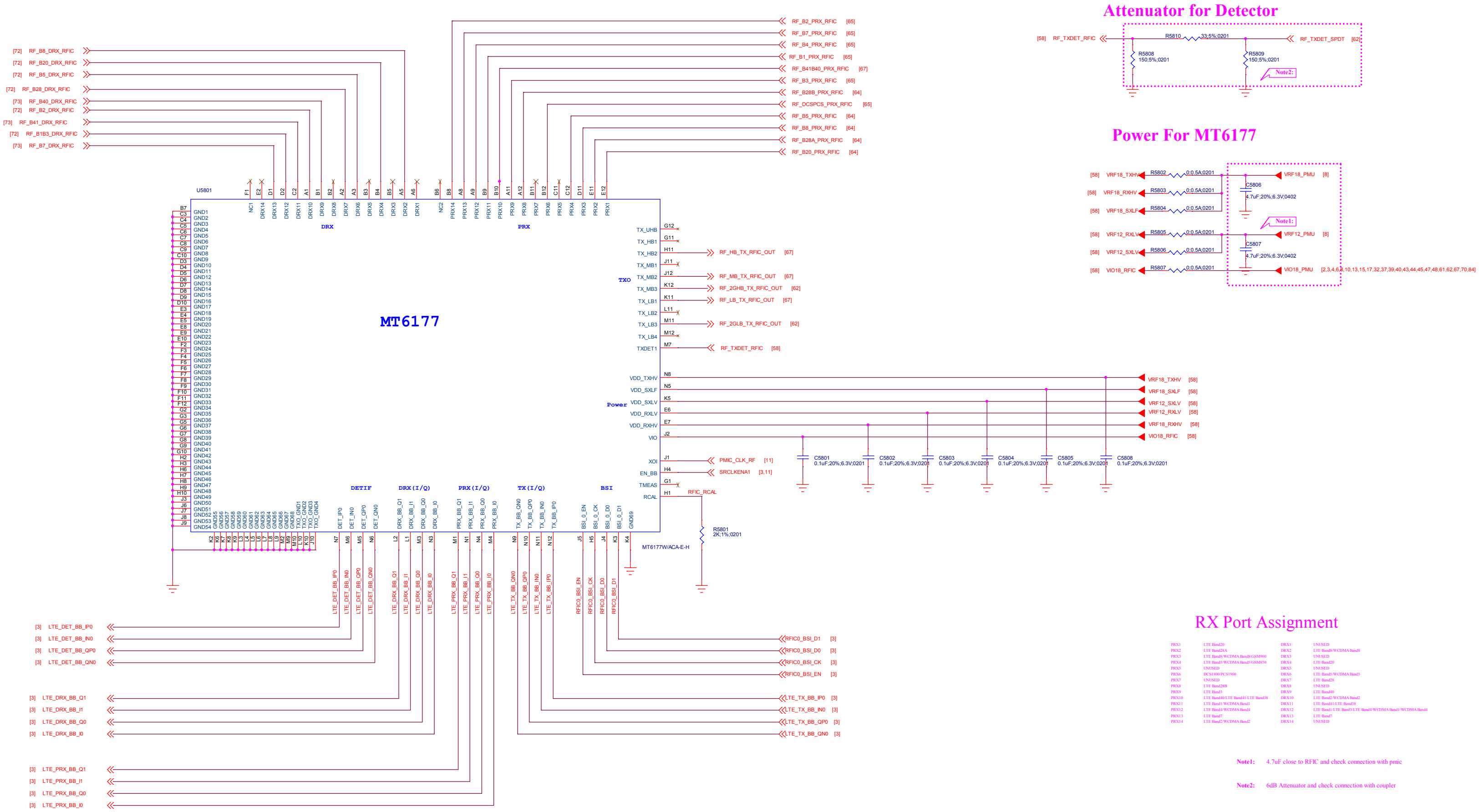
Title <Title>			
Size A1	Document Number <Doc>		Rev 1.0
Date: Thursday, May 30, 2019		Sheet 36 of 65	



RX Port Assignment

PRX1	LTE Band20	DRX1	UNUSED
PRX2	LTE Band28A	DRX2	LTE Band8/WCDMA Band8
PRX3	LTE Band8/WCDMA Band8/GSM900	DRX3	UNUSED
PRX4	LTE Band5/WCDMA Band5/GSM850	DRX4	LTE Band20
PRX5	UNUSED	DRX5	UNUSED
PRX6	DCS1800/PCS1900	DRX6	LTE Band5/WCDMA Band5
PRX7	UNUSED	DRX7	LTE Band28
PRX8	LTE Band28B	DRX8	UNUSED
PRX9	LTE Band3	DRX9	LTE Band40
PRX10	LTE Band40/LTE Band41/LTE Band38	DRX10	LTE Band2/WCDMA Band2
PRX11	LTE Band1/WCDMA Band1	DRX11	LTE Band41/LTE Band38
PRX12	LTE Band4/WCDMA Band4	DRX12	LTE Band1/LTE Band3/LTE Band4/WCDMA Band1/WCDMA Band4
PRX13	LTE Band7	DRX13	LTE Band7
PRX14	LTE Band2/WCDMA Band2	DRX14	UNUSED

Note1: 4.7uF close to RFIC and check connection with pmic

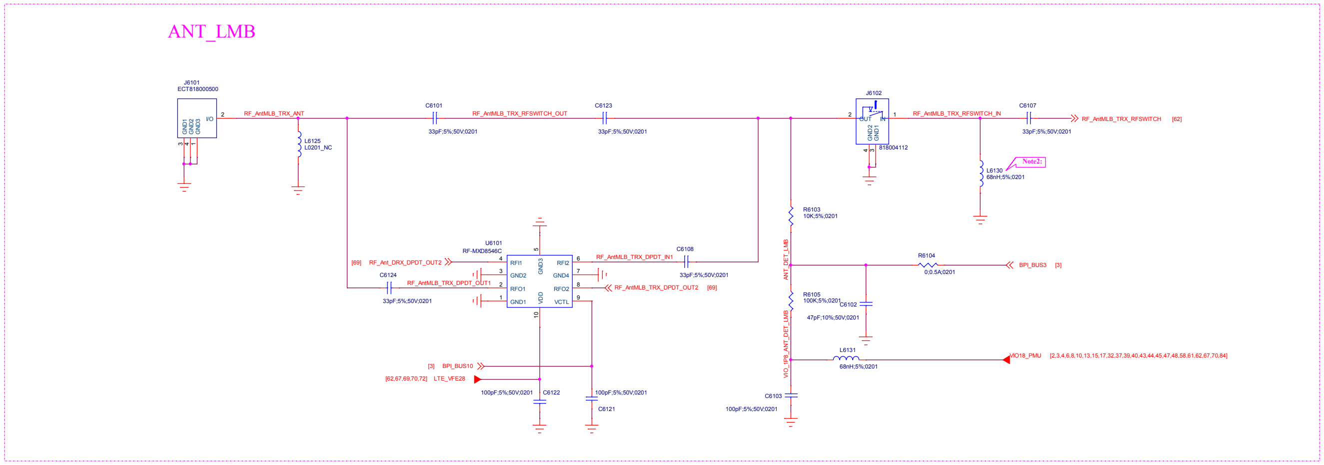
Note2: 6dB Attenuator and check connection with coupler

Title	Page Name = 58_Transceiver	REV: V10	
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530_1120	D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet 37 of 65	

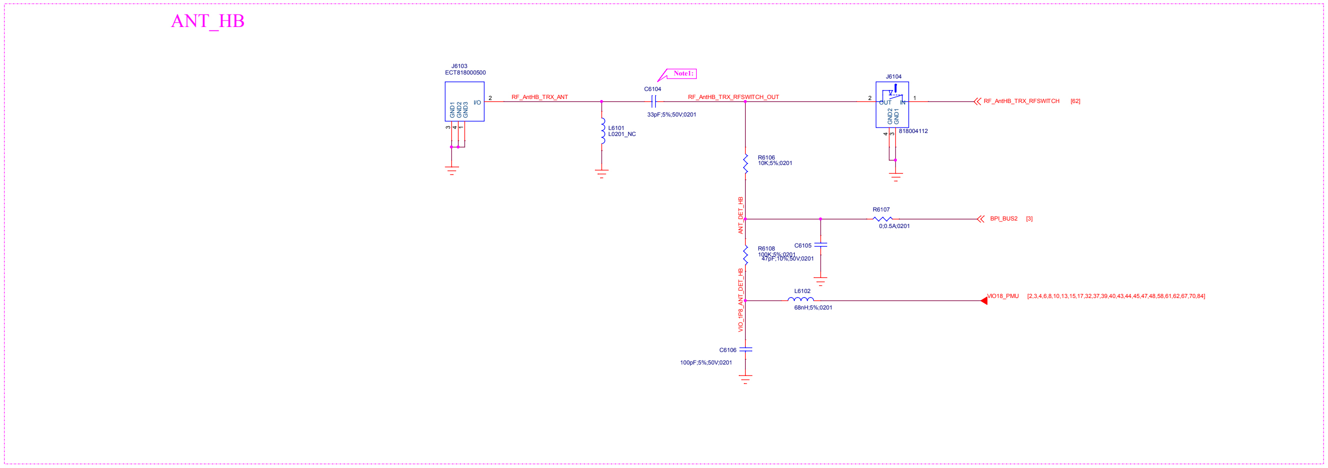
Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet	38 of 65

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet	39 of 65

ANT_LMB



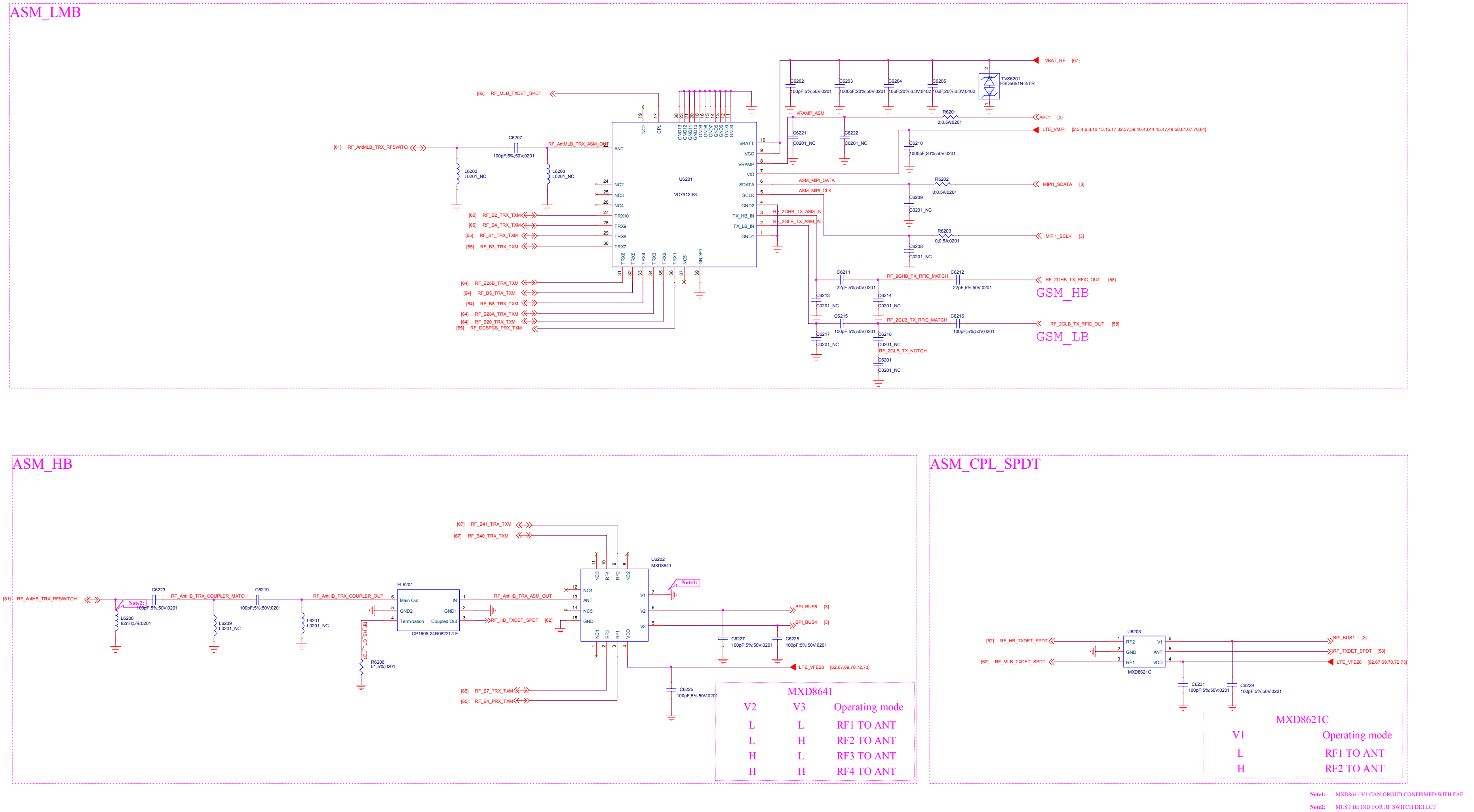
ANT_HB



Note1: RF SWITCH DETECT,MUST BE CAP IF DPDT IS WITHOUT DC BLOCK

Note2: MUST BE IND FOR RF SWITCH DETECT

Title	REV: V10
DOCUMENT NO. Design Name = 98860_1_12M13_20190530212	Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLE
WINGTECH	
Date: Page Modify Date = Thursday, May 30, 2019	Sheet 40 of 65



MXD8641

V2	V3	Operating mode
L	L	RF1 TO ANT
L	H	RF2 TO ANT
H	L	RF3 TO ANT
H	H	RF4 TO ANT

MXD8621C

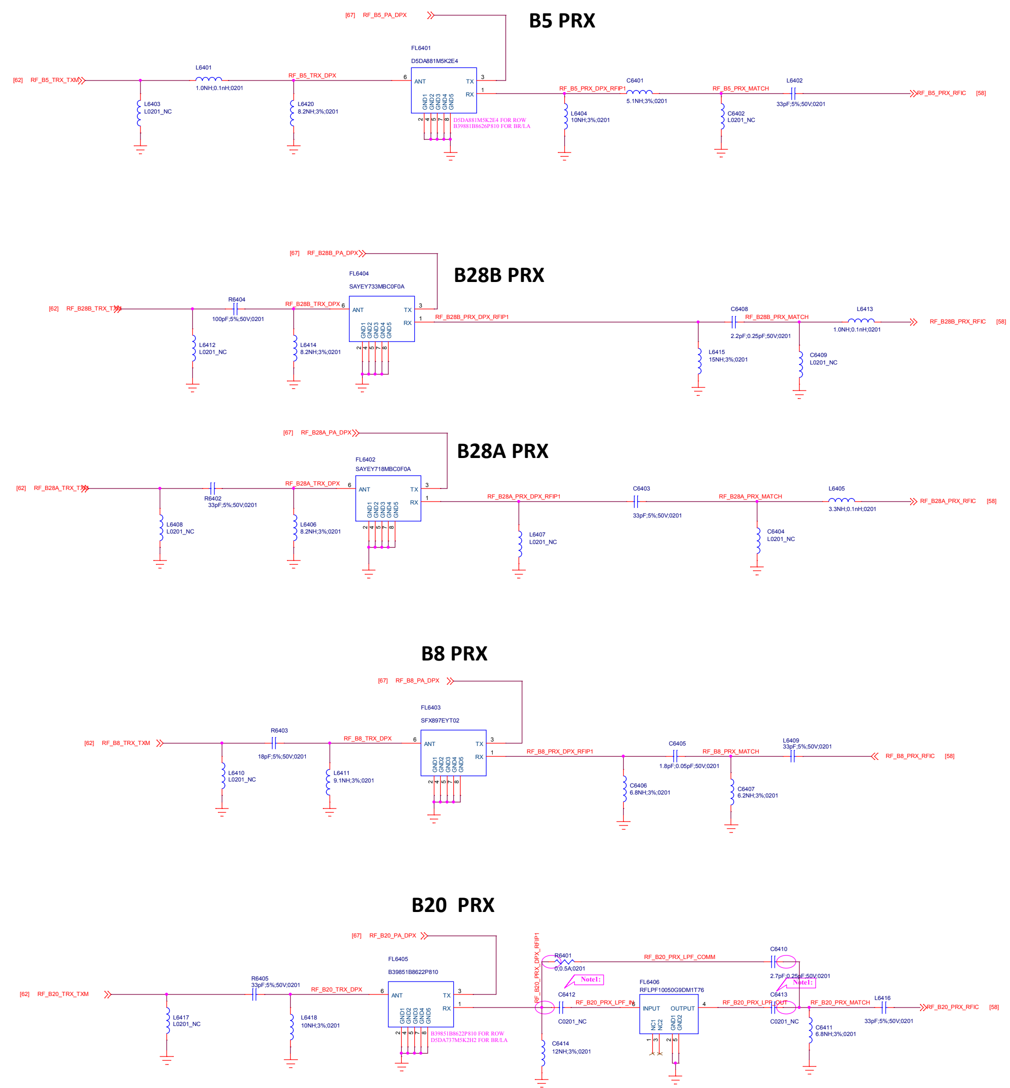
V1	Operating mode
L	RF1 TO ANT
H	RF2 TO ANT

Note1: MXD8641 V1 CAN GROUD CONFIRMED WITH FAE

Note2: MUST BE IND FOR RF SWITCH DETECT

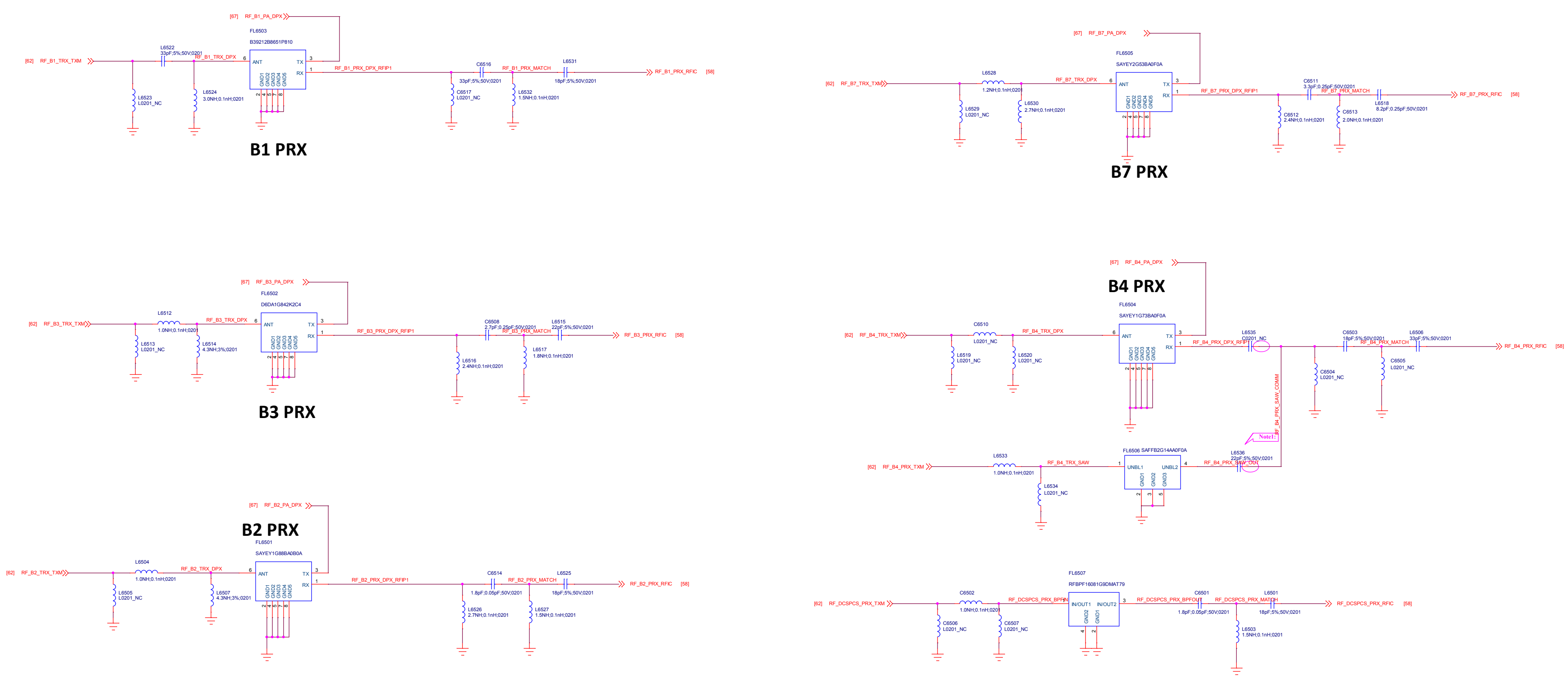
Title	Page Name = 62_Primary_ASM	REV: V10
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
WINGTECH		
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet 41 of 65

Title <Title>		
Size A1	Document Number <Doc>	Rev 1.0
Date: Thursday, May 30, 2019	Sheet 42 of 65	



Note1: COMPATIBLE DESIGN FOR LPF,AND SHARE PAD IN PCB

Title	Page Name = 64_TRX_LB	REV: V10
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530	Size D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet 43 of 65



Note1: COMPATIBLE DESIGN FOR B4 PRX SAW AND DUPLEXER,AND SHARE PAD IN PCB

Band Note:

	FL4704	FL4701	FL4707	FL4701	FL4702	FL4703	FL4702	FL4601	FL4602	FL4604	FL4603	U4802	U4805
China	B1		B3		G1900		B34/39	B5	B8			B40	B38/B41
CMCC	B1	B2	B3		G1900	B7	B34/39	B5	B8			B40	B38/B41
Asia Pacific Simple	B1		B3		G1900			B5	B8			B40	B38/B41
Asia Pacific	B1		B3		G1900	B7		B5	B8			B40	B38/B41
Overseas Full Band	B1		B3	B20	G1900	B7		B5	B8	B28A	B28B	B40	B38/B41

Title	Page Name = 65_TRX_MHB	REV: V10
DOCUMENT NO.:	Design Name = 98860_1_12M13_2019053021	Size D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet 44 of 65

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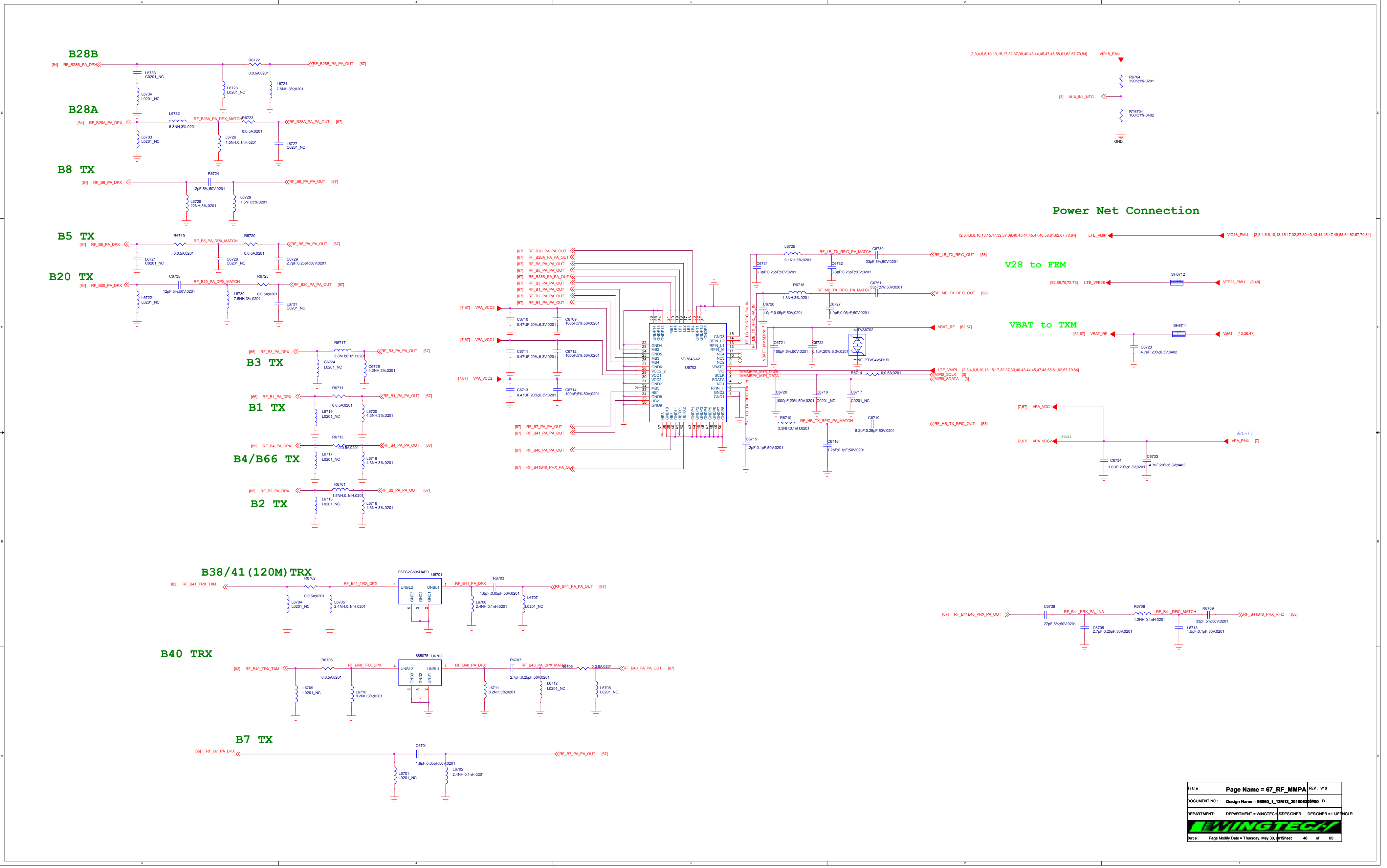
Title	<Title>				
Size: A1	Document Number: <Doc>				Rev: 1.0
Date:	Thursday, May 30, 2019	Sheet	45	of	65

5 4 3 2 1


B28B
B28A
B8 TX
B5 TX
B20 TX
B3 TX
B1 TX
B4/B66 TX
B2 TX
B38/41(120M)TRX
B40 TRX
B7 TX
Power Net Connection
V28 to FEM
VBAT to TXM
VC7643-62
U6702
F6FC2G595H4PD
U6701
885075
U6703
NF_PTVS4V5D1BL
Page Name = 67_RF_MMPA
REV: V10
Design Name = 98860_1_12M13_20190530
DEPARTMENT = WINGTECH-SZ
DESIGNER = LIUFENGLEI
Page Modify Date = Thursday, May 30, 2019
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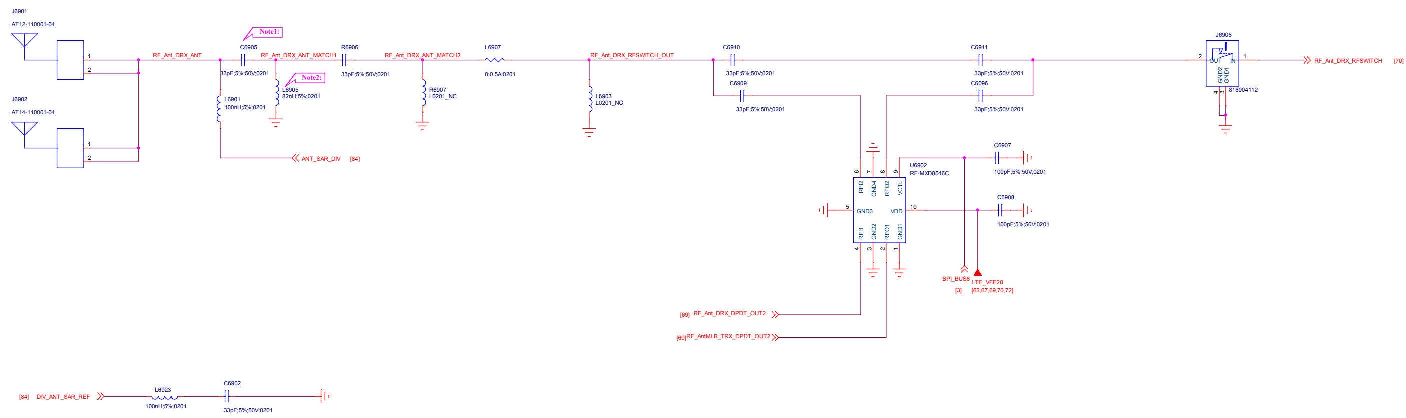
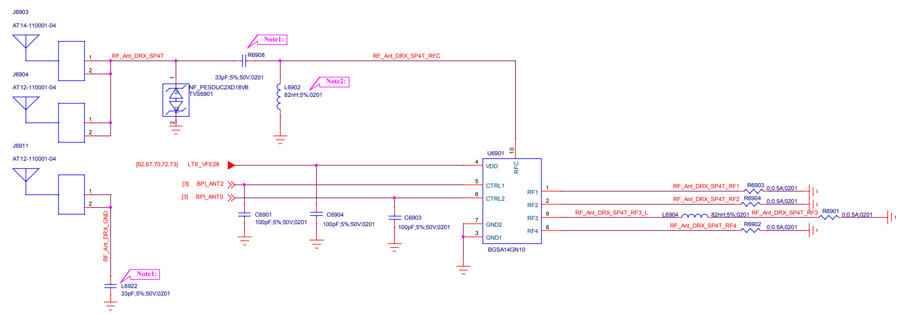
WINGTECH

Title <Title>		
Size A1	Document Number <Doc>	Rev 1.0
Date: Thursday, May 30, 2019	Sheet 47 of 65	

ANT_DIV_FEED



ANT_DIV_TUNER



BGSA14GN10

CTRL1	CTRL2	Operating mode
L	L	RF1 TO RFC
L	H	RF2 TO RFC
H	L	RF3 TO RFC
H	H	RF4 TO RFC

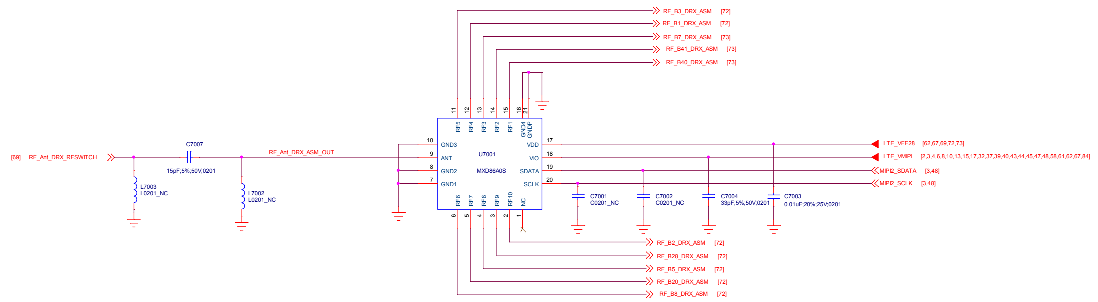
Note1: MUST BE CAP FOR SAR DETECT DESIGN

Note2: MUST BE IND FOR SAR DETECT DESIGN

Note3: 0R RES IS RESERVED FOR SAR SENSOR DESENSE,PLACE CLOSE TO ANTENNA

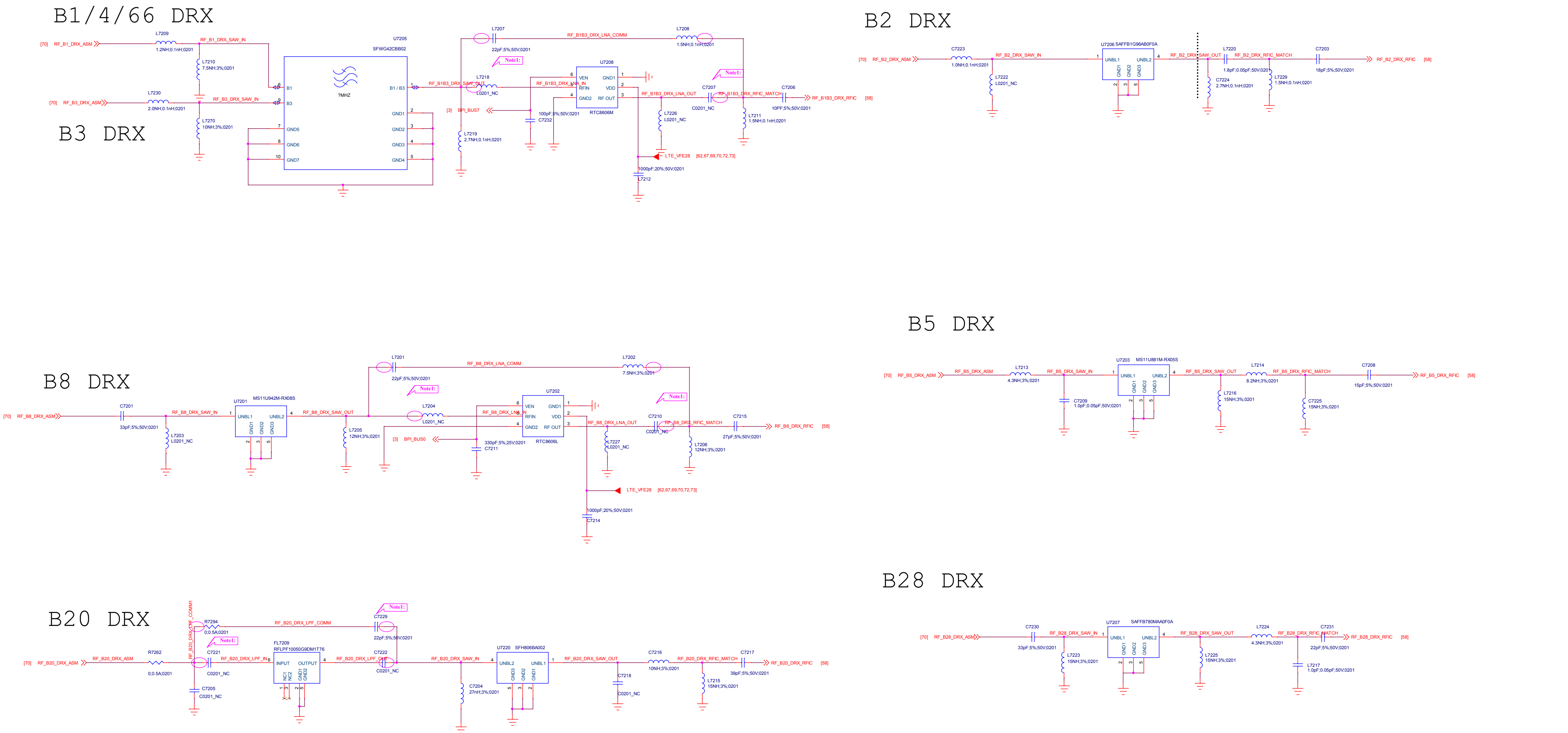
Note4: share pad in PCB

Title	Page Name = 69_Diversity_ANT	REV:	V10
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530[illegible]	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet 48	of 65



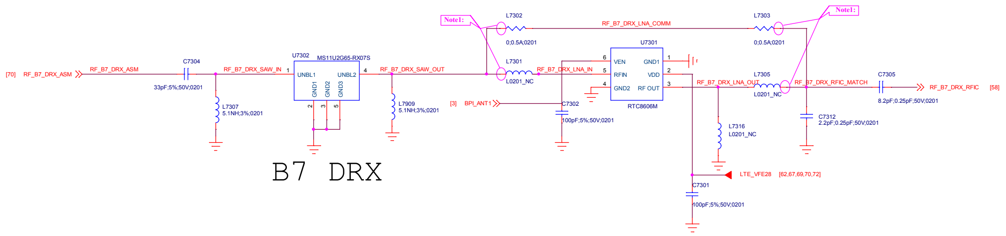
Title	Page Name = 70_Diversity ASM	Rev:	V10
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet	49 of 65

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet	50 of 65

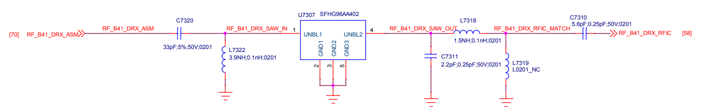


Note1: COMPATIBLE DESIGN ,AND SHARE PAD IN PCB

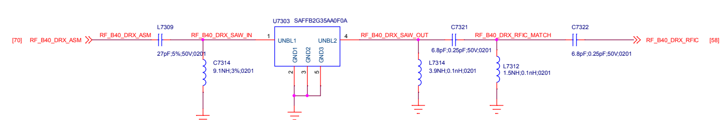
Title	Page Name = 72_DRX_LMB	REV: V10	
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530	Size D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet 51 of 65	



B7 DRX



B38/41 DRX



B40 DRX

Note1: COMPATIBLE DESIGN ,AND SHARE PAD IN PCB

Title	Page Name = 73_DRX_HB	REV: V10
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530	D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI	
WINGTECH		
Date: Page Modify Date = Thursday, May 30, 2019	Sheet 52 of 65	

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Title	<Title>	
Size A1	Document Number <Doc>	Rev 1.0
Date: Thursday, May 30, 2019	Sheet 53 of 65	

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Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 54 of 65	

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5 4 3 2 1

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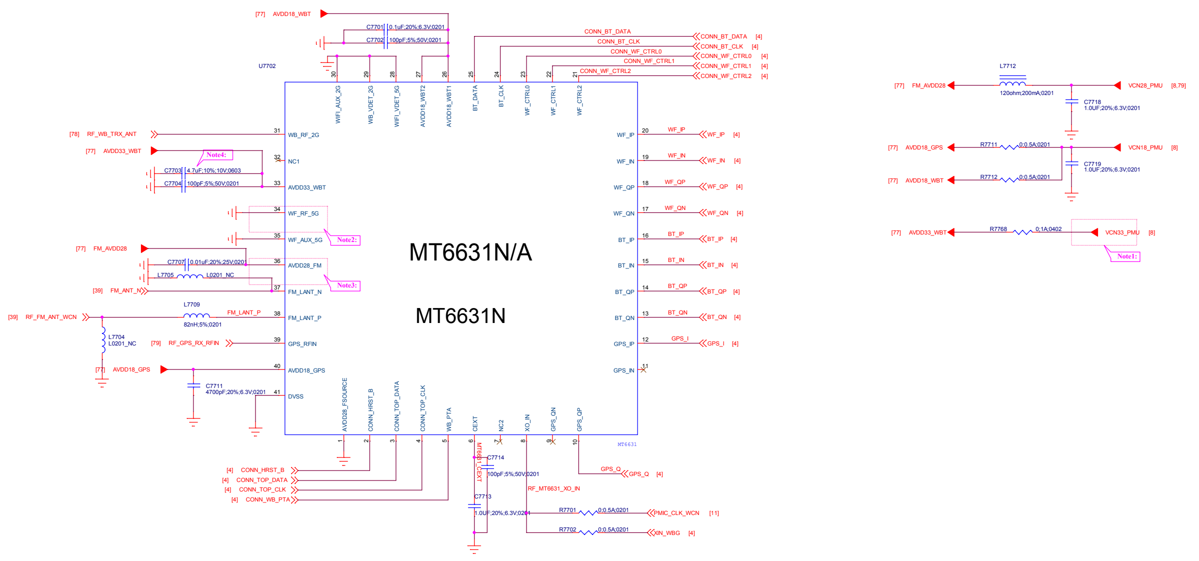
Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet	55 of 65

5 4 3 2 1



Note1: If WiFi 5G were no need, VCN33 could be chosen from PMIC output (VCN33_PMU)

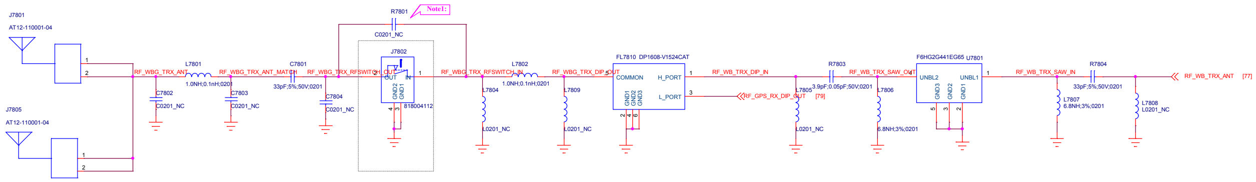
Note2: If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND

Note3: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support

Note4: 0603 package use 802167100028

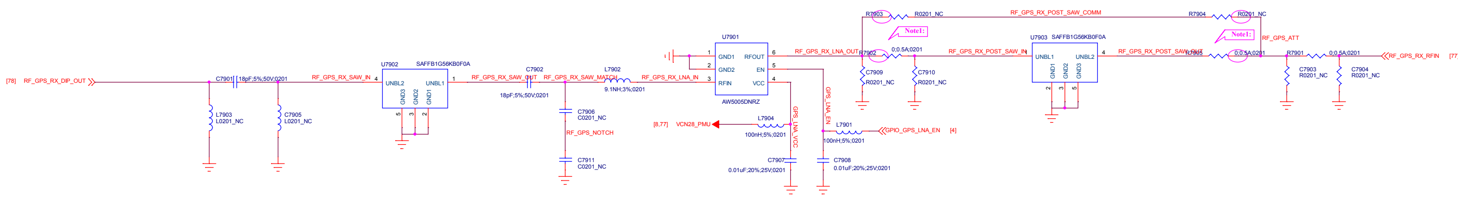
Title	Page Name = 77_WCN_MT6631	REV: V10	
DOCUMENT NO.:	Design Name = 98860_1_12M13_20190530_1120	Size D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Thursday, May 30, 2019	Sheet 56	of 65

WIFI GPS ANT



Note1: COMPATIBLE DESIGN FOR WBG RF SWITCH, USE 0R AFTER MP

Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 57 of 65	



Note1: COMPATIBLE DESIGN FOR GPS POST SAW,SHARE PAD IN PCB

Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 58	of 65

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 59	of 65

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Thursday, May 30, 2019	Sheet	60 of 65

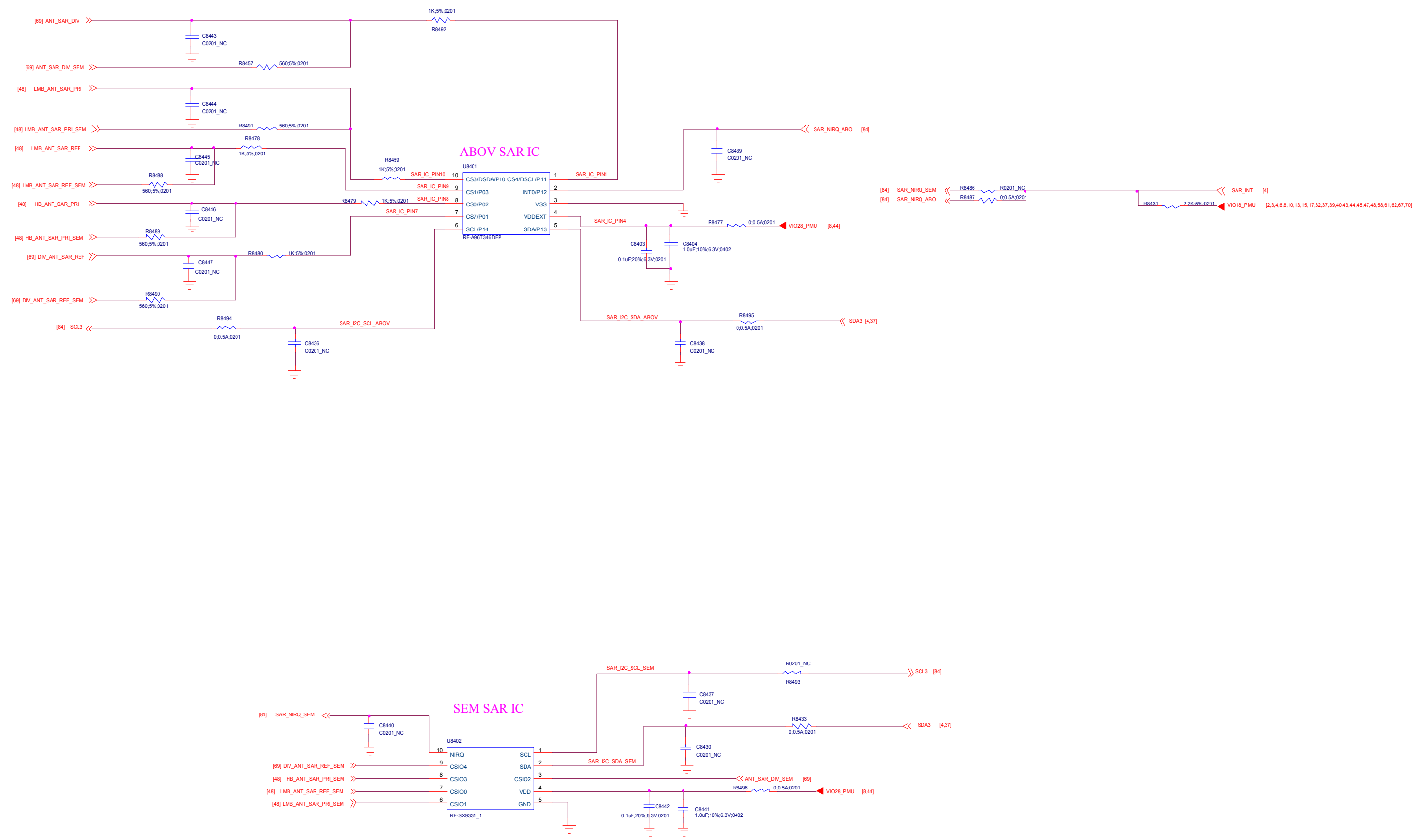
Title	10_BB_ POWER_PDN	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SZ		DESIGNER: LIUFENGLEI
Date: Thursday, May 30, 2019		Sheet 61 of 65

If SYS CLKis used, the SYS_CLK_REQ signal needs to be connected to the host GPIO pin that configures as SRCLKENAI mode and this pin must be default low

NOTE6:

C8218/C8219 place close to XIN/XOUT Balls (less than 5mm)

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet	62 of 65



Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 63 of 65	

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Thursday, May 30, 2019	Sheet	64 of 65

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, May 30, 2019	Sheet 65	of 65